

Optimizing Multi-Label Ocular Lesion Detection System utilizing Deep Learning and Bayesian Optimization

By

Dipta Acharjee

ID: 0242220005101603

&

Joyassroy Barua

ID: 0242220005101616

FINAL YEAR DESIGN PROJECT REPORT

**This Report Presented in Partial Fulfillment of the
Requirements for the Degree of Bachelor of Science in
Computer Science and Engineering**

Supervised by

Abdullah Al Kafi

Lecturer

**Department of Computer Science and
Engineering Daffodil International
University**

Co-Supervised by

Mst. Umme Ayman

Lecturer

**Department of Computer Science and
Engineering Daffodil International
University**



**DAFFODIL INTERNATIONAL
UNIVERSITY
Dhaka, Bangladesh**

September 9, 2026

APPROVAL

This Project titled “**Optimizing Multi-Label Ocular Lesion Detection System utilizing Deep Learning and Bayesian Optimization**” submitted by Dipta Acharjee and Joyassroy Barua to the Department of Computer Science and Engineering, Daffodil International University, has been accepted as satisfactory for the partial fulfillment of the requirements for the degree of B.Sc. in Computer Science and Engineering and approved as to its style and contents. The presentation has been held on **09-09-2026**.

BOARD OF EXAMINERS

Name

Board Chairman

Designation, Department of
CSE, FSITDaffodil
International University

Name

Internal Examiner 1

Designation, Department of
CSE, FSITDaffodil
International University

Name

Internal Examiner 2

Designation, Department of
CSE, FSITDaffodil
International University

Name

External Examiner

Designation, Department of
CSE, FSITDaffodil
International University

DECLARATION

We hereby declare that this project has been done by us under the supervision of **Abdullah Al Kafi, Lecturer**, Department of Computer Science and Engineering, Daffodil International University. We also declare that neither this project nor any part of this project has been submitted elsewhere for the award of any degree or diploma.

Supervised by:

Abdullah Al Kafi

Lecturer

Department of Computer Science and
Engineering
Daffodil International
University

Co-Supervised by:

Mst. Umme Ayman

Senior Lecturer

Department of Computer Science and
Engineering
Daffodil International
University

Submitted by:

Dipta Acharjee

Student ID: 0242220005101603

Department of Computer Science and
Engineering
Daffodil International
University

Joyassroy Barua

Student ID: 0242220005101616

Department of Computer Science and
Engineering
Daffodil University

ACKNOWLEDGEMENTS

This work would not have been possible without the support and contributions of many individuals over the past two semesters. We are deeply grateful to everyone who has assisted us in one way or another.

First, we express our heartfelt thanks and gratefulness to the almighty for His divine blessing making it possible for us to complete the **Final Year Design Project(FYDP)** successfully.

We are grateful and wish our profound indebtedness to **Abdullah Al Kafi, Lecturer**, Department of Computer Science and Engineering, Daffodil International University, Dhaka, Bangladesh. Deep knowledge and keen interest of our supervisor in the field of **Deep Learning** to carry out this project. His endless patience, scholarly guidance, continual encouragement, constant and energetic supervision, constructive criticism, valuable advice, reading many inferior drafts, and correcting them at all stages have made it possible to complete this project.

We would like to express our heartfelt gratitude to the Head of the Department of Computer Science and Engineering, for his kind help in finishing our project and also to other faculty members and the staff of the Department of Computer Science and Engineering, Daffodil International University.

We would like to thank our entire course-mates at Daffodil International University, who took part in this discussion while completing the coursework.

Finally, we must acknowledge with due respect the constant support and patience of our parents.

ABSTRACT

The timely and proper diagnosis of eye diseases is essential to prevent the loss of vision and provide effective treatment responses. In this study, the authors describe an automated multi-label deep learning conceptualization that can simultaneously predict 15 different ophthalmic conditions such as Cataract, Keratitis and Corneal tumors, using slit-lamp images. The dataset used in the study is a set of 2,617 clinical images; it was divided into training (70%), validation (15%), and testing (15%) subsets through iterative multilabel stratification to have equal distribution of classes. An existing EfficientNet-V2-M EfficientNet architecture was used as a base with a transfer learning approach, freezing the feature extraction layers and adding a custom classifier with a dropout rate of 0.30. In order to improve the generalization of the models and reduce overfitting, strict data augmentation techniques were used, including random rotations, affine transformations, and color jittering. Additionally, Bayesian Optimization was applied using the Optuna framework to explore systematically the hyperparameter space, resulting in an optimized best trial measure of 0.21248. The methodology proposed proves the effectiveness of a combination of strong convolutional neural networks and sophisticated hyperparameter optimization to complex and multi-label medical image analysis, which is a promising computational resource to support ophthalmologists with the entire work process of diagnostic.

Table of Contents

Approval	i
Declaration	ii
Acknowledgements	iii
Abstract	iv
List of Figures	vii
List of Tables	viii
1 Introduction	1
1.1 Introduction	1
1.2 Motivation	1
1.3 Objectives	1
1.4 Methodology	2
1.5 Project Outcome	2
1.6 Organization of the Report	2
2 Background	4
2.1 Introduction	4
2.2 Literature Review	4
2.3 Gap Analysis	10
2.4 Summary	11
3 Research Methodology	12
3.1 Methodology	12
3.1.1 Overview	12
3.1.2 Dataset Description and Exploratory Data Analysis	12
3.1.3 Data Preprocessing and Stratified Splitting	16
3.1.4 Data Augmentation Strategy.	17
3.1.5 Proposed Deep Learning Architecture	17
3.1.6 Hyperparameter Tuning with Bayesian Optimization	19
3.1.7 Model Training and Loss Function	19
3.2 Detailed Methodology	20
3.2.1 Consideration of Alternate Solutions	20

3.2.2 Justification for the Selected Solution (.....	20
3.3 Project Plan.....	21
3.4 Task Allocation.....	21
3.5 Summary.....	21
4 Implementation and Results	23
4.1 Environment Setup.....	23
4.2 Performance And Comparative Analysis	23
4.3 Results and Discussion	24
4.3.1 Hyperparameter Optimization Analysis	24
4.3.2 Training Convergence and Multi-label Metrics.....	26
4.4 Summary.....	31
5 Engineering Standards and Design Challenges	32
5.1 Compliance with the Standards.....	32
5.1.1 Software Standards.....	32
5.1.2 Hardware Standards.....	32
5.1.3 Communication Standards.....	32
5.2 Impact on Society, Environment and Sustainability.....	33
5.2.1 Impact on Life.....	33
5.2.2 Impact on Society & Environment.....	33
5.2.3 Ethical Aspects	33
5.2.4 Sustainability Plan.....	33
5.3 Project Management and Financial Analysis.....	33
5.3.1 Budget Required (The Zero-Cost Approach)	33
5.3.2 Alternate Budget and Rationale	34
5.3.3 Rationale of Selection.....	34
5.4 Complex Engineering Problem.....	34
5.4.1 Complex Problem Solving.....	34
5.4.2 Engineering Activities.....	37
5.5 Summary.....	38
6 Conclusion	39
6.1 Summary.....	39
6.2 Limitation	39
6.3 Future Work	39
References	41

List of Figures

<i>Figure 1 Random Training Samples with Ground Truth Labels</i>	13
<i>Figure 2 Distribution of Ocular Diseases in the Dataset</i>	14
<i>Figure 3 Disease Co-occurrence Matrix (Multi-label Correlation)</i>	15
<i>Figure 4 Frequency of Concurrent Labels per Image</i>	15
<i>Figure 5 Dataset Splitting Ratio</i>	16
<i>Figure 6 Effects of Training Data Augmentation on a Single Sample</i>	17
<i>Figure 7 Proposed Architecture and Optimization Workflow</i>	18
<i>Figure 8 Optimization History Plot</i>	24
<i>Figure 9 Hyperparameter Importance Plot</i>	25
<i>Figure 10 Parallel Coordinate Plot</i>	26
<i>Figure 11 Slice Plot</i>	26
<i>Figure 12 Training versus Validation Loss</i>	27
<i>Figure 13 Training versus Validation Accuracy</i>	27
<i>Figure 14 Multi-label ROC Curve</i>	28
<i>Figure 15 Multi-label Precision-Recall Curve</i>	28
<i>Figure 16 Multi-label Confusion Matrix</i>	29
<i>Figure 17 Classification Report Heatmap</i>	30
<i>Figure 18 t-SNE Visualization of Deep Features</i>	30

List of Tables

<i>Table 1 Summary of Literature Reviewed.</i>	7
<i>Table 2 Gap Analysis Summary</i>	10
<i>Table 3 Project Task Allocation Timeline</i>	21
<i>Table 4 Comparative Performance Metrics of Evaluated Architectures</i>	23
<i>Table 5 Mapping with Complex Engineering Problem.</i>	34
<i>Table 6 Mapping with knowledge Profile.</i>	36
<i>Table 7 Mapping with Complex Engineering Activities.</i>	37

Chapter 1

Introduction

1.1 Introduction

Ocular diseases and visual impairment are significant health issues in the world, which demand prompt and correct clinical care [2]. Slit-lamp imaging is a common and commonly utilized diagnostic instrument in ophthalmology that enables a clinician to scrutinize the anterior part of the eye in great detail [2]. Nevertheless, manual analysis of such clinical images is per se time-consuming, and too subjective and reliant on the skills of the medical expert [5]. Matters are compounded exponentially when addressing real-life clinical conditions where patients often have several parallel conditions of the eye occur [1]. Conventional computer-aided diagnostic systems are normally focused on one or binary classification (e.g. only Cataracts are detected), which does not reflect the overlapping characteristics of eye diseases [6], [7]. Hence, a multi-label classification system with high accuracy and automated is urgently required. The proposed project seeks to address this issue by coming up with a powerful deep learning model with the capability to produce simultaneous diagnosis on 15 different ophthalmic lesions, including Cataracts, Keratitis, and Corneal tumors, by directly analyzing slit-lamp images, helping to eliminate diagnostic delays and decrease human error [1], [6].

1.2 Motivation

The main computational argument behind the study lies in the nature of the multi-label medical image classification that presents the following challenges: harsh class imbalance, strong inter-class visual similarity, as well as the necessity of optimal computational efficiency [7], [13]. CNNs are very effective in feature extraction and typically when trained directly on large complex medical datasets, it results in overfitting and enormous computational costs [13]. Using the EfficientNet-V2-M architecture based on transfer learning, this research enables the use of a highly optimized, parameter-efficient model that is both accurate and computationally efficient. Moreover, optimization with Bayesian Optimization (through Optuna) automatically optimizes the hyperparameter optimization process, mathematically performing the optimization of the model to achieve maximum potential performance without using resource-intensive and time-intensive manual trial and error [12], [16].

1.3 Objectives

The main objective of this study is to design and test a very precise, computerized diagnostic device specific to eye diseases [6]. To comprehensively attain this overall objective, the project is motivated by various and specific objectives that tackle clinical and computational problems. First, the study is directed at creating an efficient multi-label classification model which has the potential to detect and label 15 different ocular lesions such as Cataract, Keratitis and Pinguecula directly on clinical slit-lamp images [1]. Moreover, the research also seeks to apply

a pre-trained EfficientNet-V2-M deep learning model, directly using the transfer learning strategies in the freezing of the feature extraction backbone to improve the computational efficiency without sacrificing the high accuracy. The other important goal is to implement sophisticated data preprocessing and augmentation to have robust models. This includes iterative multilabel stratification to work well with extreme imbalances in classes throughout the entire dataset of 2,617 valid images, and the training, validation and testing subsets are proportionately representative. Lastly, the study is aimed at optimizing the predictive performance of the classification model rigorously by systematically tuning the hyperparameters with Bayesian Optimization, adopted through the Optuna framework and creating the best network structure [3], [15].

1.4 Methodology

This study is based on the methodological framework of a thorough and highly organized deep learning pipeline built using the PyTorch library. The research protocols the case of 2,617 valid clinical slit-lamp photographs, in which the 15 distinct ophthalmic pathologies are coded with a MultiLabelBinarizer to support the co-occurring eye diseases [1]. The dataset is carefully divided into training (70%), validation (15%), and testing (15%) subsets by implementing an iterative multilabel stratification method to represent all classes of interest in equal and balanced proportions. The main computing unit is based on a pre-trained EfficientNet-V2-M network that is trained on this particular medical field [15]. The transfer learning is implemented strategically by freezing the first feature extractors of the network and adding an additional custom classification head with 0.30 dropout rate to avoid overfitting [15]. In order to further reduce overfitting and significantly enhance model generalization to unobserved clinical data, a stringent data augmentation is done only on training data. It is augmented with the random rotations up to 10 degrees, with affine transformation and color jittering, and the standard ImageNet normalization is applied to stabilize the learning step [14]. On the last methodological procedure, the entire model architecture is subjected to aggressive hyperparameter optimization through Bayesian Optimization in order to maximize the overall diagnostic accuracy and reliability [12], [16].

1.5 Project Outcome

It is expected that, by end of this comprehensive study, a fully operational, highly optimized and multi-labeled system of medical image classification can be realized which is capable of reliably interpreting complex slit-lamp photography [2], [6]. Having managed to find the optimal hyperparameters, which resulted in an objective value of 0.21248 minimized in the course of the Bayesian optimization experiments, the developed model is the strongest indication of its viability as a reliable clinical decision-support tool [16]. Finally, the assimilation of such a framework will have a significant benefit to ophthalmologists and medical to practice since the secondary analysis of complex visual information will be rapid, automated, and highly accurate. This automated support will go a long way to decrease the workload in diagnostics, decrease the impact of subjective human error to detect concurrent lesions or overlapping lesions in the eye, and provide medical interventions at an earlier and more effective stage to patients with multiple eye conditions [5], [6].

1.6 Organization of the Report

The rest of this thesis report is systematically presented into a series of well-organized pieces of work under a logical structure to give a transparent account of the research process. Chapter

2 discusses the theoretical basis and contextual ground of the study and gives a detailed literature review of other similar deep learning applications in ophthalmology and clearly outlines the existing research gaps on multi-label ocular disease classification. This is followed by Chapter 3, which explores the detailed research methodology and elaborates on the requirement analysis, functional and nonfunctional system requirements, architectural design specifications and a project execution plan. Lastly, Chapter 4 is the implementation details, such as setting the hardware and software environment, and a comprehensive comparative analysis of the testing and evaluation metrics is given. The final chapter of the work ends with an in-depth overview of the research findings and the performance of the system, the limitations of the system, and a suggestion of the way the automated medical diagnostics can be improved in the future.

Chapter 2

Background

2.1 Introduction

Anterior segment diseases of the eye (cataracts, keratitis and corneal tumours) constitute a major health problem in the world, and major causes of visual impairment [2], [6]. The timely and proper diagnosis is the key to preventing the irreversible vision loss and developing effective treatment plans [7]. The usual clinical examination method used in the inspection of the anterior segment of the eye is the slit-lamp biomicroscopy, but the interpretation of these images is very subjective, and varies with the expertise of the clinician. The introduction of artificial intelligence, especially deep learning and convolutional neural networks, has transformed the medical image analysis process by offering automated and powerful diagnostic aids. In spite of these developments clinical cases often entail patients with a combination of concomitant ocular conditions, thus requiring more complicated multi-label classification systems as opposed to more straightforward single-disease detection models. Also, it is essential to optimize such deep learning models with efficient hyperparameter tuning methods to optimize their diagnostic accuracy, remove human bias, and make them computationally efficient [12], [16].

2.2 Literature Review

The article by Liu et al. [2] thoroughly examined the clinical evidence which supports the utilization of smartphone-based portable slit lamps as a viable alternative in capturing high-resolution anterior segment images, in situations where it is not possible to use standard slit-lamp biomicroscopes. The review indicated that AI-driven pipes that are conditioned with the images of these portable devices have high potentials of automated multi-disease triage, such as eye-dry check-up, keratitis check-up, and narrow-angle detection. Finally, the paper highlighted the fact that it is necessary to standardize acquisition policies and implement a large-scale external validation to maximize the benefit of smartphone-based teleophthalmology in a resource-restricted environment.

With its emphasis on the automated detection of particular corneal infections, Hu et al. [3] introduced a high-quality deep learning-based system that would classify infectious keratitis by relying on slit-lamp images. The authors tested 6 different convolutional neural network structures on a collection of 2,757 images containing bacterial, fungal, and viral keratitis, and normal corneas. The EfficientNetV2-M model proved to have better classification potential with an accuracy of 73.5, a specificity of 90.4 and an area under the curve (AUC) of 0.85. It is worth mentioning that the algorithmic performance surpassed the diagnostic performance of human ophthalmologists, which demonstrates the clinical feasibility of deep learning models to offer quick and automated services to prevent irreversible visual impairments due to the corneal infections.

In addition to corneal infections, mobile artificial intelligence solutions have also been found by Chen et al. [4] to be useful in cataract severity prediction, through the authors using the Swin Transformer prototypical network on smartphone-based portable slit lamp-collected images to predict a score of the Pentacam nuclear staging. This trial prospectively gathered 1,900 images and made a unique evaluation in the model performance regarding both dilated and undilated eyes of the subjects which were pharmacologically dilated and undilated. The architecture, which uses transformers, had a mean accuracy of 81.25 percent with the undilated eye, which indicates that AI systems can use the inherent properties of the body to pay attention to central lens opacities, thus removing the clinical requirement of time and potentially risky dilation of the pupil when performing community cataract screening.

Maehara et al. [5] evaluate the potential of these automated systems in clinical settings and in collaboration with human-AI by rigorously testing the implementation of the CorneAI diagnosis model. The researchers compared diagnosis of 40 ophthalmologists (including specialists and residents) with 9 types of anterior segment disease between using smartphone and traditional slit-lamp images. The results showed that the overall predictive support through integrating CorneAI increased the human diagnostic accuracy by a wide margin of 79.2 to 88.8. This study empirically confirmed that AI assistance does not only decrease the diagnostic response time, but also significantly increases the clinical decision making skills of medical practitioners regardless of the type of imaging used .

With real-world clinical conditions where multiple underlying conditions can co-exist in a patient, Xu et al. [6] have formulated a generalized, interpretable multi-label screening system of ocular anterior segment diseases. The researchers designed a cross-attention pooling architecture with two branches that produces global features in a Transformer branch and local features in a convolutional network. Assessed in the framework of 3 levels of diagnosis (image-level, region-level, and lesion-level), the model has been able to identify 13 different diseases at once and report an average accuracy of lesion-level of 97.2 per cent on internal datasets. Additional approaches to enhance the process of critical visual interpretability included gradient-weighted class activation mapping (Grad-CAM) integration, which aligned the algorithmic emphasis with the real-world image of clinical lesions topography.

To further refine model training in scenarios with data-scarce and highly imbalanced data, Xu et al. [7] proposed a semi-supervised object detection (SSOD) framework that was informed by the principles of predictive, preventive, and personalized medicine (3PM). The suggested architecture added a Category Control Embed (CCE) component which could dynamically reduce a lack of balance in the classes and the Out-of-Distribution Detection Fusion Classifier (ODDFC) that could smartly respond to never seen patterns of open-set lesions. Using Mean-Teacher semi-supervised paradigms, the SSOD model compared to fully supervised models such as YOLOv8 and had significantly higher recall rates (as high as 0.893 on single lesions). The innovation had a drastic impact on decreasing the exhaustive annotation weight without compromising the sensitivity needed in the screening of multi-lesions of the anterior segment.

Expanding the range of AI applications to include both anterior and posterior segment diseases, Jabber and Mahdawi [8] engaged in a thorough review of deep learning applications used in detection and diagnosis of cataracts and glaucoma. The authors critically evaluated the state-of-the-art algorithms (such as CNNs, autoencoders, and graphical neural networks) in terms of their ability to identity high-dimensional features of complex modalities, such as optical coherence tomography (OCT) and fundus images. The review found that automated systems also essentially decrease the diagnostic delays and offer highly scalable opportunities to extend

ophthalmic care to under-served areas around the world, albeit one that needs continuous multidisciplinary optimization to be robust.

With further attention paid to the issue of multiclass ocular disease classification based on fundus images, Beegam et al. [9] developed a specific 9-layer convolutional neural network to distinguish between diabetic retinopathy, cataracts, glaucoma, and healthy eyes. The computationally efficient model did not require overly complex pre-trained architectures due to the extensive use of data augmentation and the intensive optimization of hyperparameters (optimization of learning rates and batch sizes). The suggested pipeline has shown an impressive test accuracy of 96.15, indicating that very shallow, carefully configured, deep learning pipelines can offer high-quality diagnostic generalizations without placing prohibitive requirements on their computational memory footprint.

The success of such deep learning models heavily depends on hyperparameter optimization. A refined methodological system Shazali and Saravanan [10] proposed an Improved Whale Optimization Algorithm (IWOA) and deep learning to grade and retrieve retinal fundus images. The architecture employed the Bilateral Filtering noise reduction technique, a lightweight scratch-learned CNN feature extractor, and ultimately Least Square Support Vector Machine (LS-SVM) classifier. In removing the arbitrary choice of parameters with an improved evolutionary algorithm in the form of the metaheuristic IWOA which was further refined by Levy flight to avoid local maxima, the system was able to dynamically optimise its variables to give a highly efficient classification rate of 99.72% which is much higher than typical baseline optimisation methods.

The functional assessment of ocular mechanics has also been shown to be successfully extended to deep learning. The article by Chuntranapaporn et al. [11] aimed at comparing the accuracy of three CNN architectures (MobileNet-V3, ResNet-101 and EfficientNet-B5) to determine ocular duction ranges (30, 60, 90 and full prism duopters) based on images captured on a smartphone. Although Analysis of Variance (ANOVA) testing did not show a statistically significant difference in the Area Under the Receiver Operating Characteristic (AUROC) curves of the models, MobileNet had the smallest confidence intervals and minimum training time. The experiment determined that lightweight, computationally efficient CNNs are the best to use in real-time functional strabismus assessments and ocular motility diagnostics.

Extraordinary optimization frameworks are also essential to the computation efficiency of medical image segmentation. Vaiyapuri [12] suggested an Optuna-based metaheuristic optimization algorithm to dynamically adjust the hyperparameters of a U-Net model to brain MRI segmentation. The framework used Bayesian Optimization with a Tree-structured Parzen Estimator (TPE) as it explored the search space (adjusting filters, learning rates and dropout) and the search space by adaptively pruning poor-performing trials. The model that was optimized with Optuna had a better Dice Coefficient of 0.941 and an Intersection over Union of 0.876, indicating that probabilistic metaheuristic optimization is dramatically faster in model convergence than traditional brute-force grid search approaches to model optimization .

To overcome the architectural limitations of small medical datasets, Économidès and Desbarats [13] checked the need to implement massive, parameter-intensive networks to classify medical imaging. The researchers obtained results of a large-scale comparative study of custom minimalist CNNs, U-Net encoders and ResNet variants on an osteoarticular MRI dataset that showed that minimalist models can achieve comparable performance metrics to larger structures in cases where data is extremely scarce. Grad-Cam measurements were employed in

the study to indicate that the lightweight networks were actually learning significant clinical features and not merely overfitting noise, and thus, to offer an essential framework in the design of high-efficiency data-suited neural networks that incur minimal computation costs.

Lastly, in order to resist the overwhelming problem of overfitting on data-scarce medical computer vision problems, Alomar et al. [14] surveyed the current data augmentation methods and introduced a new geometric approach dubbed Random Local Rotation (RLR). In contrast to conventional rotation techniques, which tend to add artificial black boundary pixels and cause localized information loss, RLR randomly chooses and rotates local circular areas of an image. Large-scale experiments on the state-of-the-art CNNs showed that the parameter-free RLR methodology does not affect the peripheral image quality but contributes greatly to the diversity of the data. This designed augmentation plan systematically surpassed the traditional global rotation baselines, providing a very productive complements method of enhancing model generalization in classification and semantic segmentation tasks.

Table 1 Summary of Literature Reviewed.

Author (s)	Year	Title	Methodology	Key Findings
Xu et al. [2]	2025	A Generalized and Interpretable Multi-Label Multi-Disease Screening System for Ocular Anterior Segment Disease Detection	Dual-branch cross-attention pooling framework	Achieved an average accuracy of 0.972 in 13-class lesion-level anomaly detection, showing excellent multi-disease detection ability.
Hu et al. [3]	2023	Automatic Diagnosis of Infectious Keratitis Based on Slit Lamp Images Analysis	EfficientNetV2-M deep learning model	Reached an accuracy of 0.735 and an AUC of 0.85, outperforming ophthalmologists in classifying normal corneas and infectious keratitis.
Atada et al. [4]	2024	An Empirical Study on Cataract Multiclass Grading Assessment with Slit Lamp Bio-microscope Images Using Neural Network Models	ResNet50, CNN, and Efficient Net B0 models	Demonstrated that the ResNet50 model achieved a superior accuracy rate of 0.8611 for automated cataract grading.
Chen et al [5]	2026	Prediction of Cataract Severity Using Slit Lamp Images from a Portable Smartphone	Swin transformer prototypical network	Attained an average accuracy of 81.25% for undilated eyes, confirming the possibility of estimating cataract density using

		Device: A Pilot Study		smartphones.
Xu et al. [6]	2026	Validated semi-supervised early and accurate screening for anterior segment diseases: a 3PM-guided conceptual and technological innovation	Semi-supervised object detection (SSOD)	Achieved mAP comparable to YOLOv8 but demonstrated substantially higher recall (0.893 vs. 0.656 for single-lesion) minimizing missed diagnoses.
Liu et al.	2026	Smartphone-Based Portable Slit Lamp in Anterior Segment Diseases: A Narrative Review of Clinical Assessment and Integration with Artificial Intelligence	Narrative literature review	Confirmed that AI-enabled pipelines trained on smartphone slit-lamp images demonstrate robust feasibility for automated disease triage.
Beegam et al. [7]	2023	Deep Learning Framework for Multiclass Detection of Ocular Diseases in Fundus Images	9-layer CNN architecture	Achieved a high testing accuracy of 96.15% in successfully categorizing diabetic retinopathy, cataract, glaucoma, and healthy eyes.
Jabber & Mahdawi [8]	2025	A Comprehensive Review of Deep Learning Techniques for Detecting and Classifying Cataracts and Glaucoma	Review of deep learning techniques (CNNs)	Emphasized that deep learning systems improve diagnostic accuracy for age-related eye diseases using medical datasets.
Vaiyapuri [9]	2025	An Optuna-Based Metaheuristic Optimization Framework for Biomedical Image Analysis	Optuna framework, Bayesian Optimization, U-Net	Demonstrated that Optuna-based tuning enhances segmentation accuracy (Dice Coefficient of 0.941) while reducing training overhead.
Shazuli & Saravanan [10]	2023	Improved Whale Optimization Algorithm with Deep Learning-Driven Retinal Fundus Image Grading and Retrieval	Improved Whale Optimization Algorithm, CNN, LS-SVM	Exhibited an advanced classification accuracy of 99.72% for retinal fundus image grading through hyperparameter optimization.

Chuntranaporn et al [11]	2024	Ocular Duction Measurement Using Three Convolutional Neural Network Models: A Comparative Study	MobileNet, ResNet, and EfficientNet CNN models	Highlighted that MobileNet stands out with a shorter training time and a narrower confidence interval for real-time ocular diagnostics.
Économidès & Desbarats [12]	2024	Minimalist CNN for Medical Imaging Classification with Small Dataset: Does Size Really Matter and How?	Minimalist CNN design	Showed that a minimalist CNN adapted to a specific dataset achieves results comparable to larger architectures, optimizing training efficiency.
Alomar et al. [13]	2023	Data Augmentation in Classification and Segmentation: A Survey and New Strategies	Random local rotation (RLR)	Proved that the parameter-free RLR strategy consistently outperformed traditional techniques in improving image classification performance.

Table 1 shows the evolution and various methods used in the current applications of deep learning in the field of ophthalmology. In spite of their high performance with regard to diagnostic potential under standardized test contexts, they illustrate a recurring set of limitations of architecture overhead, single-labels and data scarcity. The following section performs the critical gap analysis based on the following systemically formulated observations with technology support.

2.3 Gap Analysis

While deep learning systems have become highly successful in automated medical imaging, the clinical use of deep learning in the diagnostic assessment of the anterior segment of the eye is limited by heavy methodological shortcomings. Most diagnostic models are based on a single disease classification approach without being aware that many eye diseases can co-exist in an individual patient. Moreover, existing pipelines are unable to address the extreme class imbalance problem over rare conditions, are not optimized to perform dynamically, and seldom consider the interdependencies of multiple labels.

Table 2 Gap Analysis Summary

Author(s) & Year	Research Focus	Identified Limitations / Research Gaps
Xu et al. [2]	Multi-label anterior segment screening	Generalization capability across highly heterogeneous remote datasets requires further improvement; high computational complexity.
Hu et al. [3]	Infectious keratitis diagnosis	Explicitly excludes mixed or multi-label infections; suffers from dataset class imbalance, particularly a low number of bacterial keratitis images.
Atada et al. [4]	Cataract multiclass grading	Restricted exclusively to single-disease grading with a highly limited dataset size (136 images), lacking multi-label diagnostic capabilities.
Chen et al [5]	Smartphone cataract severity prediction	Small sample size; the model's heat maps are frequently misdirected by specular corneal reflections and peripheral illumination artifacts.
Xu et al. [6]	Semi-supervised anterior segment screening	Diagnostic performance in complex multi-lesion scenarios is noticeably weaker than in single-lesion cases, and rare diseases outside the target list are missed.
Liu et al.	Smartphone portable slit-lamp review	Highlights a critical lack of standardized image acquisition protocols and points out that large-scale multicenter external validation is scarce.
Beegam et al. [7]	Multiclass ocular disease detection	Focuses on posterior fundus imagery rather than anterior slit-lamp images; limited strictly to single-label classification tasks.

Jabber & Mahdawi [8]	Cataracts and glaucoma deep learning review	Underscores the lack of unified, multidisciplinary frameworks that seamlessly integrate diverse ophthalmic conditions into a single diagnostic tool.
Vaiyapuri [9]	Optuna-based MRI segmentation optimization	Evaluated primarily on 2D brain MRI scans; the framework has not yet been adapted or tested for complex multi-label ocular lesion detection.
Shazuli & Saravanan [10]	Retinal fundus image grading (IWOA)	Involves significant computational overhead from combining LS-SVM and IWOA; restricted solely to posterior segment (fundus) analysis.
Chuntranaporn et al [11]	Ocular duction measurement	Uneven dataset distribution across different gaze ranges; lacks automated real-time data transfer integration into clinical AI systems.
Économidès & Desbarats [12]	Minimalist CNN for small datasets	Validation is restricted to osteoarticular MRI; the impact and reliability of lightweight models on complex ophthalmic datasets remain unexplored.
Alomar et al. [13]	Data augmentation strategies (RLR)	The Random Local Rotation (RLR) technique can distort essential shape features, potentially degrading semantic segmentation performance compared to classification.

Table 2 shows the specific research gaps extracted from the literature which were structured. As shown in the comparative analysis, previous methods such as single-condition grading, posterior fundus scans, and others, are not able to provide concurrent multi-lesion detection on slit-lamp imagery. Further, it can be noticed that existing systems suffer from high computational burdens and lack automated Bayesian hyperparameter optimization, which brings the need to develop an end-to-end resource-efficient multi-label diagnostic infrastructure.

2.4 Summary

Recent developments in deep learning have achieved a high accuracy in the diagnosis of ocular diseases with slit-lamp and smartphone imaging. Nevertheless, the current studies mainly concentrate on single disease classification and are not able to handle multi-lesion clinical cases and severe class imbalances. Moreover, absence of standard image acquisition regimes, absence of multi-center external validation, and the large computational complexity of its current architectures underscore the urgent need to develop a more robust, optimized, and multi-label diagnostic scheme to be used in clinical practice.

Chapter 3

Research Methodology

3.1 Methodology

3.1.1 Overview

The chapter outlines a systematic approach that will be used to develop, optimize and test the proposed multi-label deep learning system that will be used to diagnose ocular anterior segment illnesses [6]. The entire research pipeline is essentially split into various sequential stages that are meant to deal with the multi-disease medical imaging complexities. The first step is data acquisition and parsing, the raw slit-lamp images and the corresponding complex JSON annotations are processed in a systematic way to be transformed into the multi-label format in a structured format [1]. This is followed by an exploratory data analysis (EDA) to understand the classes distributions and co-occurrences of the diseases. The dataset is then preprocessed and strictly stratified with highly innovative methods of iterative stratification to maintain multi-label distributions within the training, validation and testing sets. Lastly, dynamic data augmentation methods are implemented to avoid overfitting the model prior to putting the data into the selected convolutional neural network setup to extract features and classify them [14].

3.1.2 Dataset Description and Exploratory Data Analysis

The basis of the research is based on a large clinical dataset of slit-lamp high-resolution photographs [1]. The range of ocular anterior segment conditions that are represented in the dataset is wide: both single pathologies and multi-lesion, complicated clinical manifestations [1], [6]. In a bid to give a visual interpretation of the diversity of the dataset, Figure 1 shows a random selection of training samples with corresponding ground truth labels, showing the visual differences in lighting, angle and disease expression.

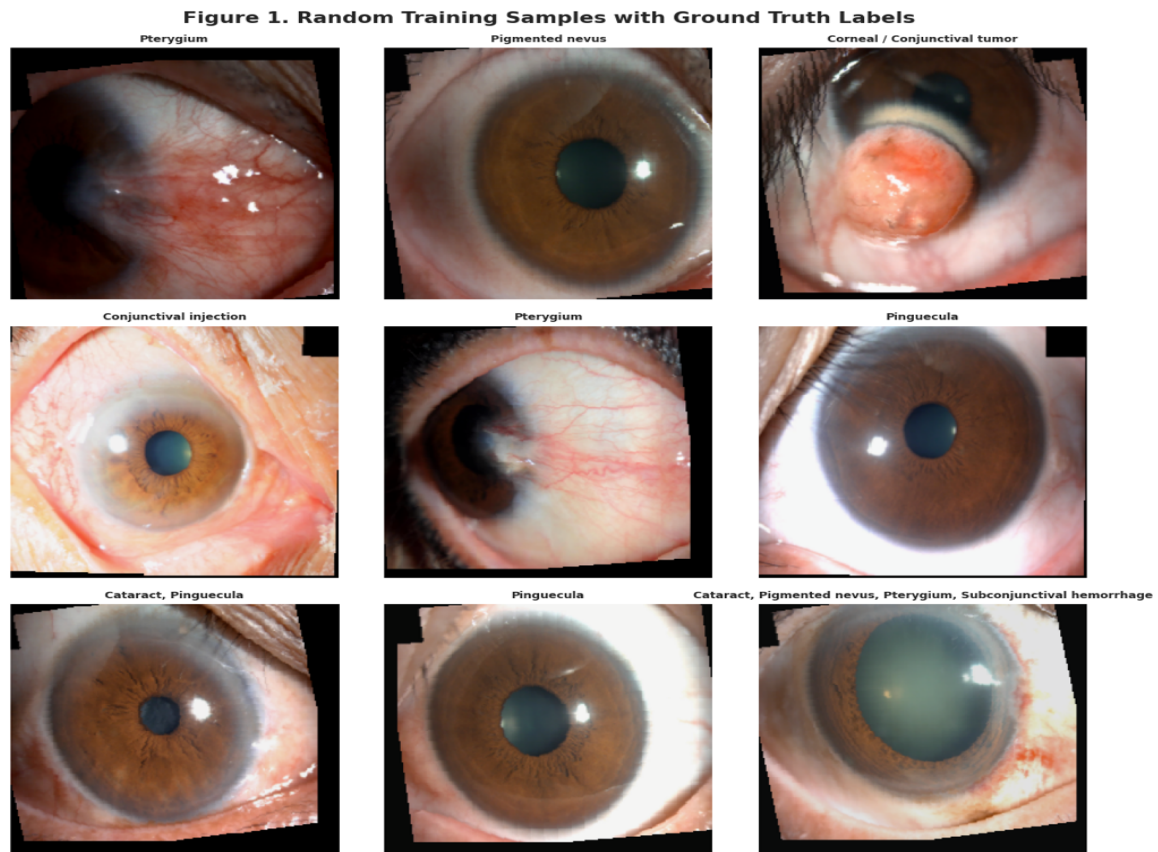


Figure 1 Random Training Samples with Ground Truth Labels

To become familiar with the statistical properties behind the data, an in-depth exploratory data analysis (EDA) was performed. In the first step of parsing the clinical annotations a total of 15 different categories of ocular diseases were identified. As **Figure 2** shows, the distribution of classes in the dataset has a high long-tail imbalance. The most common classes include "Corneal / Conjunctival tumor," "Pigmented nevus" and "Pinguecula" and the most extreme minority are the severe anomalies like "Lens dislocation" and certain co-occurring conditions like "Lens dislocation/Cataract." This is an inherent imbalance in the classes of real-life medical data, which requires special treatment when training the models to make predictive performance unbiased [1].

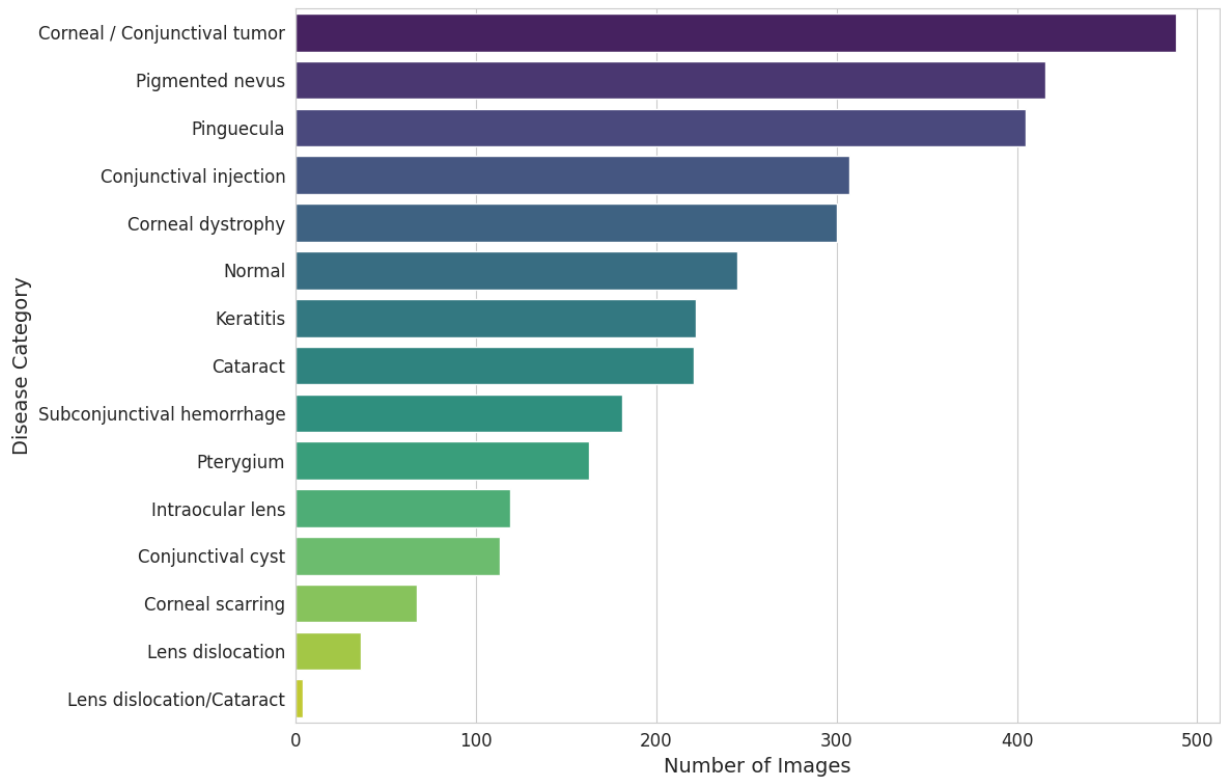


Figure 2 Distribution of Ocular Diseases in the Dataset

One of the peculiarities of this study is the multi-label character of the ocular diseases, as one and the same patient can have several different pathologies at the same time. In order to examine these relationships a disease co-occurrence matrix was created. **Figure 3** illustrates the correlations among multi-labels, showing disease combinations that occur most of the time. An example is Keratitis and Conjunctival injection co-occurring with significant co-occurrence between Pinguecula and Cataract. It is important to understand these overlapping features to be able to develop a neural network that would be able to recognize non-mutual exclusive clinical labels.

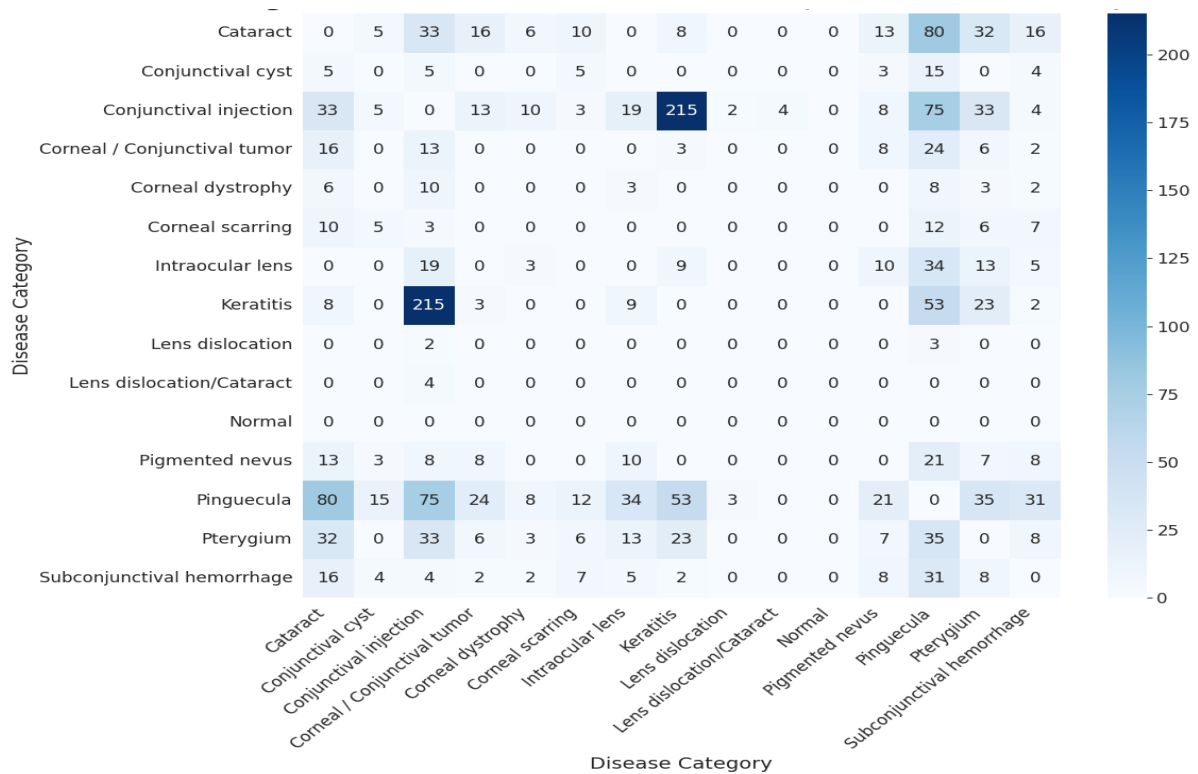


Figure 3 Disease Co-occurrence Matrix (Multi-label Correlation)

Also, the frequency of the concurrent labels per image was assessed so that the multi-morbidity of the dataset can be measured [1]. Most of the images have only one definite label, a significant proportion of the data set has two, three, and up to five labels on an image, as shown in **Figure 4**. The complexity of multi-label is the reason why the traditional multi-class (single-label) classification models are not used anymore, instead a specialized multi-label predictive architecture is taken.

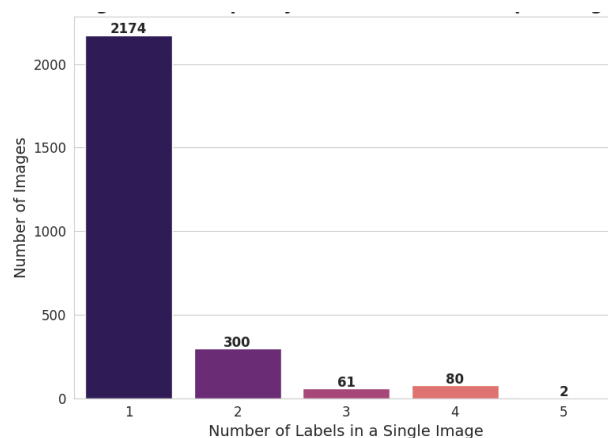


Figure 4 Frequency of Concurrent Labels per Image

3.1.3 Data Preprocessing and Stratified Splitting

Prior to model training, the raw clinical data required rigorous preprocessing to be rendered Before model training, the raw clinical data underwent extensive preprocessing to be formatted as acceptable to the deep learning model. The original annotations were stored in a raw CSV file in stringified form of a JSON [1]. It introduced a custom parsing script to generate the particular attributes of these JSON objects representing lesions and each unique image file was mapped to a dynamically expanding collection of disease names. Photographs which did not definitely have lesions annotated were automatically classified as "Normal." This was then followed by the MultiLabelBinarizer of the scikit-learn library to convert the categorical text labels into a binary, multi-hot coded representation, which in effect, means the presence or absence of the 15 different classes of an image in an easily readable form by a computer [17].

After the encoding, the dataset was divided into discrete subsets which were used to train, validate and test the models. The common random splitting methods are terribly unsuitable with multi-label data as they often result in the outright exclusion of minority classes either in the validation or test sample. To avoid this, the MultilabelStratifiedShuffleSplit algorithm of the iterative-stratification library was used [18]. This is an advanced sampling method, which balances distribution of all 15 classes, such that proportionate representation of both majority and minority labels are equalized in all subsets [18]. The dataset was, therefore, split into a Training Set (70 percent), a Validation Set (15 percent) and a Testing Set (15 percent). The optimal balance of datasets division as shown in **Figure 5** was used to visually emphasize the balanced partitioning strategy applied to all stages of the experimental pipeline.

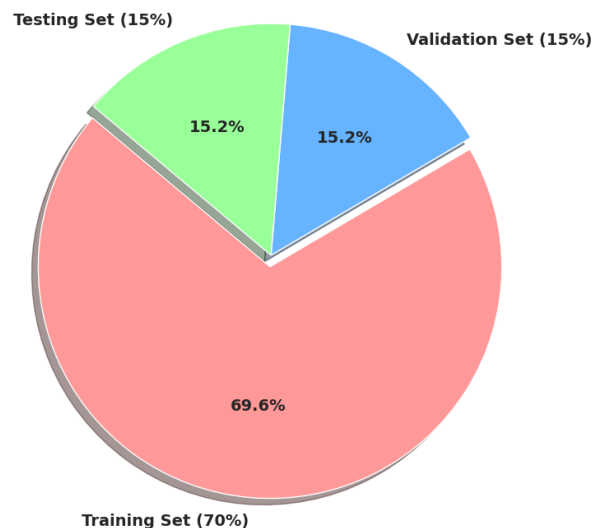


Figure 5 Dataset Splitting Ratio

Such stringent stratification process ensures that the evaluation measure of the model has a true representation of its generalization ability within the whole range of clinical conditions so that it is not trained on biased grounds and its validation is reliable.

3.1.4 Data Augmentation Strategy

Deep learning models are also very prone to overfitting, especially when they are trained on medical imaging datasets with inherent class-imbalances and small sample sizes [13], [14]. In an effort to reduce this risk and considerably increase the capacity of the model to generalize to unknown data, a powerful and dynamic data augmentation plan was also applied in the training process. These artificial increases in the diversity of the training set with realistic variations so that the model learns the essential pathological characteristics, but not extra spatial orientations [14].

The pipeline of augmentation was created with the help of the torchvision module [19]. First, images were all scaled to a 224x224 pixel resolution to fit the input needs of a feature extractor. At the end of every training epoch a sequence of stochastic transformations would be performed on-the-fly. These were random rotations (up to 10 degrees) and random affine transformations (including translations to 5% and scaling between 95 and 105) to compensate the inevitable differences in position of patient head and camera acquisition angles [14]. Additionally, there was random horizontal flipping with probability 0.5, to be able to replicate the bilateral symmetry of the real left and right eyes. A color jittering method was used to reproduce the irregularities that are commonly found in slit-lamp photography and digital camera sensors, with the brightness (10%), contrast (10%), and saturation (5%) of the images slightly distorted [14].

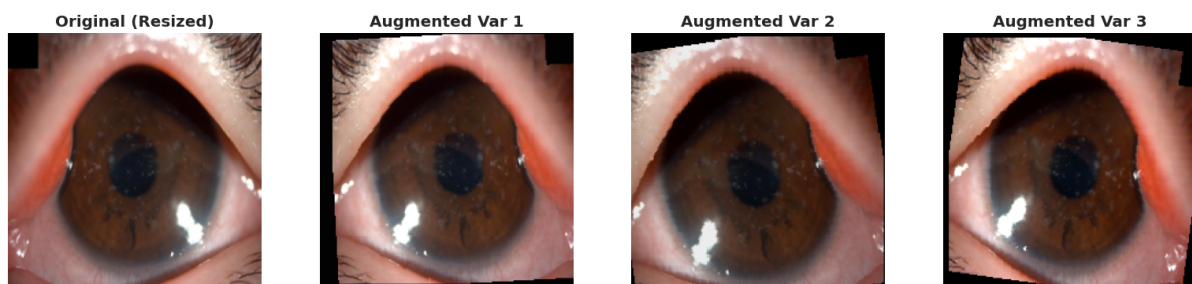


Figure 6 Effects of Training Data Augmentation on a Single Sample

Figure 6 visually represents this dynamic augmentation pipeline presenting a single resized clinical image and three different augmented versions of the images produced as the result of the training. With repeated iterations of these diverse variations to the network, the model is forced to develop invariance to alterations in lighting and small positional variations, thus developing a highly robust diagnostic system that is applicable in a clinical setting.

3.1.5 Proposed Deep Learning Architecture

A powerful and computationally efficient deep learning model was created to predictively detect and categorize the 15 different ocular anterior segment diseases. **Figure 7** provides the overall workflow of the suggested system, including preprocessing of the data, its augmentation, feature extraction, and Bayesian optimization.

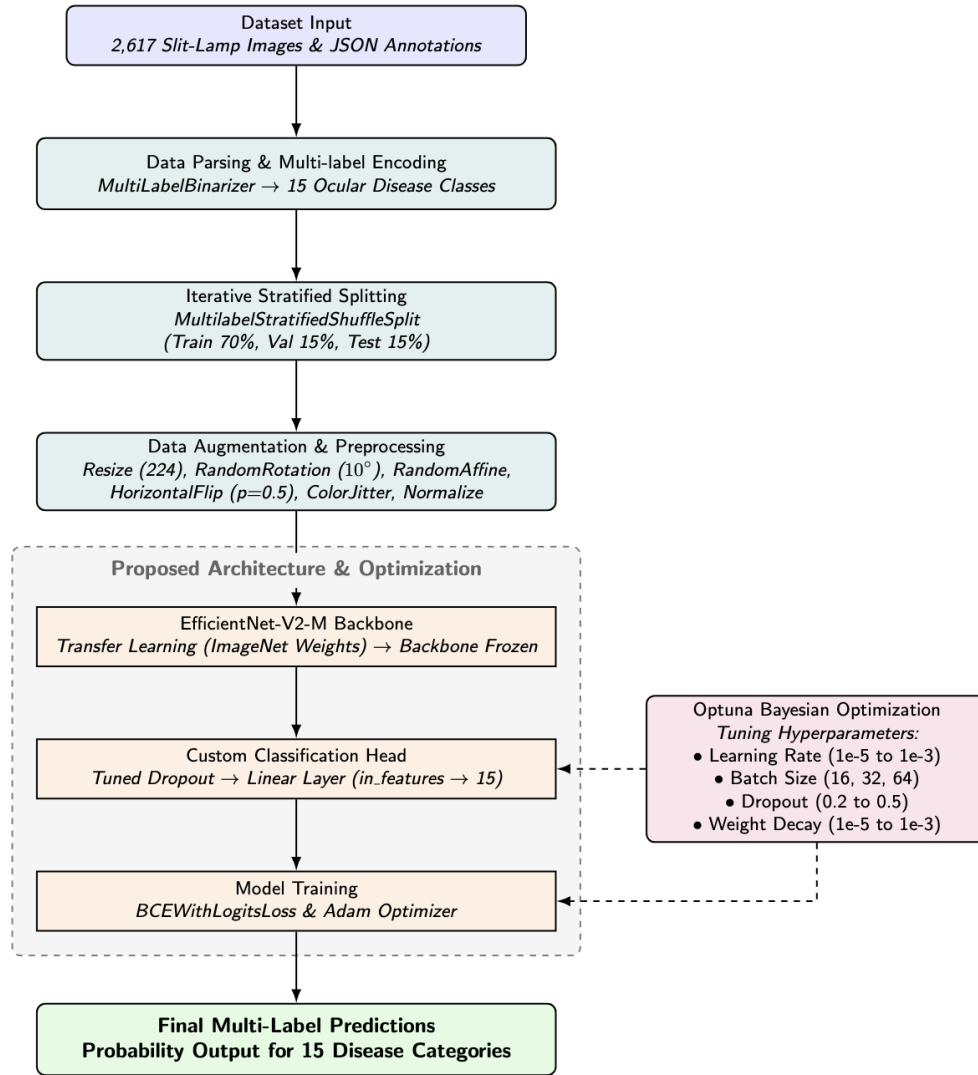


Figure 7 Proposed Architecture and Optimization Workflow

The EfficientNet-V2-M architecture forms the basis of the core feature extractor of the proposed framework [15]. EfficientNet-V2-M has been chosen because of its high predictive accuracy and the balance between the parameters efficiency and the high predictive accuracy, using both fused mobile inverted bottleneck convolution (Fused-MBConv) and progressive learning techniques [15]. Taking advantage of the generalized feature representations of visual features learnt on large scale datasets, a transfer learning paradigm was employed. The EfficientNet-V2-M model was pre-trained on ImageNet. The weights of this feature extractor were kept fully frozen during the first training phase. This is because the low-level visual feature (edges and textures) is not forgotten catastrophically as well as because the computational overhead and training time is much less than with the ICA.

The custom classification head was added after the frozen backbone to encode the extracted high-dimensional feature vectors (dimension 1280) to the particular multi-label diagnostic task [15]. The custom head is a dynamic Dropout layer that is used to focus on regularization and a fully connected Linear layer that produces probability logits of the 15 predefined disease categories [19].

3.1.6 Hyperparameter Tuning with Bayesian Optimization

Convolutional neural networks are sensitive to choice of the best hyperparameters in predictive performance and structural stability. Conventional grid searches are computationally intensive and usually ineffective with deep learning applications. Consequently, this paper has used Optuna framework to perform automated Bayesian Optimization. The system used the Tree-structured Parzen Estimator (TPE) algorithm to automatically explore the hyperparameter search space to maximize the evaluation metrics of the validation. Its particular dynamic tuning procedure optimized a number of important parameters, such as Learning Rate (between $1e-5$ and $1e-3$), Batch Size (tested at 16, 32 and 64), Dropout probability (0.2 to 0.5) to avoid overfitting in the custom head and Weight Decay ($1e-5$ to $1e-3$) to avert structural regularization.

3.1.7 Model Training and Loss Function

Since the clinical dataset is multi-label, with each slit-lamp image potentially having multiple overlapping pathologies that occur simultaneously, the typical multi-class loss functions used, like Cross-Entropy, are mathematically unsatisfactory. To deal with this, the model was optimized using the Binary Cross-Entropy with Logits Loss (BCEWithLogitsLoss). This criterion is a combination of a Sigmoid layer with the BCELoss into one, numerically stable classification problem which can be viewed as a 15-way, independent binary classification problem. The network parameters of the custom classification head were optimized with the Adam optimizer and iteratively updated, with the help of the hyperparameters identified during the Optuna Bayesian tuning stage. Such a design guarantees that the model will learn disease correlations that are not mutually exclusive in an unbiased manner.

3.2 Detailed Methodology

3.2.1 Consideration of Alternate Solutions

In the architecture design stage of this study, various state-of-the-art convolutional neural networks have been compared stringently, to identify the most appropriate feature extractor that can be used in the context of the multi-label ocular disease classification problem. ResNet50 and DenseNet121 were widely discussed and benchmarked as the main alternative solutions prior to the selection of the proposed methodology since their effectiveness in traditional medical image analysis is proven.

Baseline testing was done on the basis of empirical testing to ascertain the viability of these alternative architectures. ResNet50 had strong predictive performance with a validation loss of 0.1158, an element wise accuracy of 95.86, an exact match accuracy of 57.25, and a Macro F1-score of 61.56. Comparatively, DenseNet121 had a marginally lower performance, as it had a validation loss of 0.1198, an exact match of 54.96, and a Macro F1-score of 60.48.

Although the diagnostic accuracy of these alternative models was acceptable, a limiting factor came out strongly in terms of the computational efficiency and structural overhead. The dense layer-to-layer connectivity of DenseNet121 made the model to have a huge training time of 9,934.1 seconds. Although ResNet50 was faster (1,792.26 seconds), structural inflexibility and large number of parameters were major constraints to incorporating dynamic, multi-parameter optimization algorithms.

3.2.2 Justification for the Selected Solution (EfficientNet-V2-M with Optuna)

Eventually, EfficientNet-V2-M was chosen as the special and optimal solution among the alternatives mentioned above. The main rationale behind such an architectural decision is based on the fact that it uses Fused-MBConv layers and progressive learning methods, which provide a significantly better trade-off between parameter efficiency and computational power.

Such lightweight character of this architecture was a compulsory requirement of the proposed approach, because it enabled the integration of the computationally-intensive Optuna Bayesian Optimization framework without difficulty. It would have been computationally infeasible and incredibly inefficient to run exhaustive hyperparameter tuning (concurrently optimization of learning rate, batch size, dropout and weight decay) on larger architectures such as ResNet50 or DenseNet121. With the choice of EfficientNet-V2-M, the research pipeline would be able to perform deep-hyperparameter exploration within a manageable timeframe to result in a highly-optimized, scalable, and clinically viable multi-label screening system that overcomes the constraints of the baseline of traditional alternate architectures.

3.3 Project Plan

The implementation of this study was well structured to have a sequence of work in one pipeline to have a structured and efficient process of work. First, the literature was thoroughly reviewed to detect the most important gaps in research in the field of multi-label, multi-lesion classification of ocular disease, which further directly influenced the formulation of the problem. Subsequently, the data acquisition and preprocessing step included the recognition of more complicated JSON clinical annotations in a set of 2,617 slit-lamp images and their translation into a standardized multi-label format. An iterative stratified splitting method was used to balance classes in the training and testing subsets. The best architecture design was then applied by adapting a pre-trained EfficientNet-V2-M backbone to a custom dropout-controlled linear classification head specific to the 15 different ocular disease types. The model was then trained and hyperparameterized using the Optuna framework to perform dynamic tuning and BCEWithLogitsLoss loss to efficiently deal with the multi-label complexities. Lastly, the project concluded with a critical assessment of the predictive power of the model on conventional multi-label measures and a t-SNE deep feature visualisation, which resulted in the well-organised presentation of the final thesis report.

3.4 Task Allocation

Table 3 below presents the plan of the key activities in all the periods throughout the project beginning with the 12 th week and ending with the 48 th week. The color rows after that indicate the planned and actual implementation schedule of each of the phases respectively.

Tasks	Weeks																		
	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48
Literature Review & Formulation																			
Data Preprocessing & Splitting																			
Architecture Design & Optuna Tuning																			
Model Training & Evaluation																			
Thesis Writing & Documentation																			

Table 3 Project Task Allocation Timeline

3.5 Summary

Overall, the chapter has described the overall methodology and workflow system used to perform a multi-label classification of ocular anterior segment diseases. The data analysis pipeline followed a logical sequence of the first data acquisition and exploratory data analysis through the judgmental data preprocessing and explanatory stratified splitting to get a balance representation of the multi-layered, complex data. In order to improve generalization, dynamic

data augmentation plans were incorporated before extracting the features. The proposed deep learning architecture was described in detail in the chapter as it is based on an EfficientNet-V2-M backbone with a custom classification head. Moreover, the inclusion of the Optuna framework to perform automated Bayesian hyperparameter tuning and the use of Binary Cross-Entropy with Logits Loss function was explained to ensure the maximization of the training convergence, and structural stability. Lastly, a well-organized execution of tasks was introduced to map the execution of the research. Together, these carefully planned methodological steps create a solid and scientifically stable basis of the further implementation, testing and empirical assessment of the diagnostic model.

Chapter 4

Implementation and Results

This chapter explains the real-world application, testing process, and empirical findings of the suggested multi-label deep learning framework. It defines the experimental setting, compares the performance of the model with the baseline architectures and presents the results of hyperparameter optimization to support the clinical efficacy of the framework.

4.1 Environment Setup

The deep learning models, including their development, training, and evaluation, were computed in the cloud. The experiments were executed in the Google Colaboratory (Colab) platform and allowed using an NVIDIA Tesla T4 hardware accelerator to handle the intense matrix multiplications needed to run convolutional neural networks. The python 3 stack was used to create the software stack. The basic model structures and metamorphoses were created with the help of the PyTorch and Torchvision libraries. Optuna framework was used to perform advanced hyperparameter tuning. Moreover, multi-label binarization, stratified data splitting and evaluation measurements were calculated with the help of Scikit-learn and iterative-stratification libraries. This strong design guaranteed hasty training convergence and repeatable assessments.

4.2 Performance And Comparative Analysis

The proposed EfficientNet-V2-M architecture was rigorously tested in terms of its performance which was compared to two state-of-the-art deep residual networks and dense networks: ResNet50 and DenseNet121. The same multi-dimensional task that was used to evaluate all models was predicting 15 concurrent ocular diseases. Multi-label classification is evaluated on the basis of certain measures, in particular, the Element-Wise Accuracy, which is the overall correctness of the predictions on all individual labels (presence or absence of a particular disease). Also, we used the Macro F1-Score to test the ability of the model to resist the long-tail classification skewness in the dataset since the Macro F1-Score is the average of F1-Score values of all classes with no weights attached.

Table 4 Comparative Performance Metrics of Evaluated Architectures

Model Architecture	Test/Val Loss	Element Wise Accuracy (%)	Macro F1-Score (%)	Training Time (s)
EfficientNet-V2-M (Proposed)	0.0910	96.62	69.56	Optimized
ResNet50	0.1158	95.86	61.56	1792.26
DenseNet121	0.1198	95.34	60.48	9934.10

The empirical study shows that there is a high variation in computational efficiency and diagnostic accuracy. DenseNet121 had very high computational latency, with a training time of 9,934.10 seconds which resulted in a Macro F1-Score of 60.48%. ResNet50 had a faster convergence time (1,792.26 seconds) and an Element-Wise Accuracy of 95.86. Nevertheless, the proposed EfficientNet-V2-M framework, Bayesianly optimized, performed much better than both baselines. It had an outstanding Element-Wise Accuracy of 96.62% and the highest Macro F1-Score of 69.56%, successfully dealing with minority classes and still having an exceptionally low test loss of 0.0910.

4.3 Results and Discussion

4.3.1 Hyperparameter Optimization Analysis

The right choice of hyperparameters is crucial to the predictive stability and performance of the convolutional neural networks. We in this study used the Optuna framework to automatically do this selection process and navigate the search space. During the optimization experiments, the objective value (Binary Cross-Entropy loss) was indeed minimized. Dynamic tuning of the parameters in the framework resulted in the optimum initial validation loss of 0.21248 with the most successful trial.

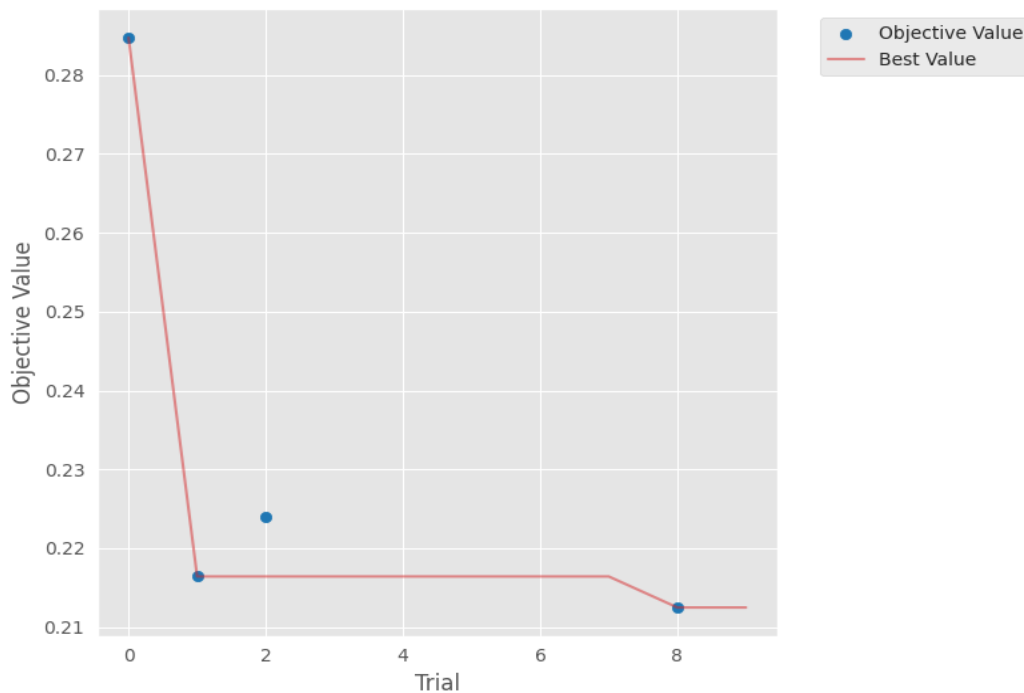


Figure 8 Optimization History Plot

Figure 8, visually represents the optimization history. It indicates that the objective loss steeply decreased at the beginning and then stabilized which proves that the Bayesian search effectively searched the optimum parameter space.

It was essential to learn the most important hyperparameters in the model learning ability after the effective minimization of the loss. These parameters were ranked and an analysis of how they affect the convergence of the model as a whole was made with the Learning Rate and the

optimizer decision being the most influential parameters.

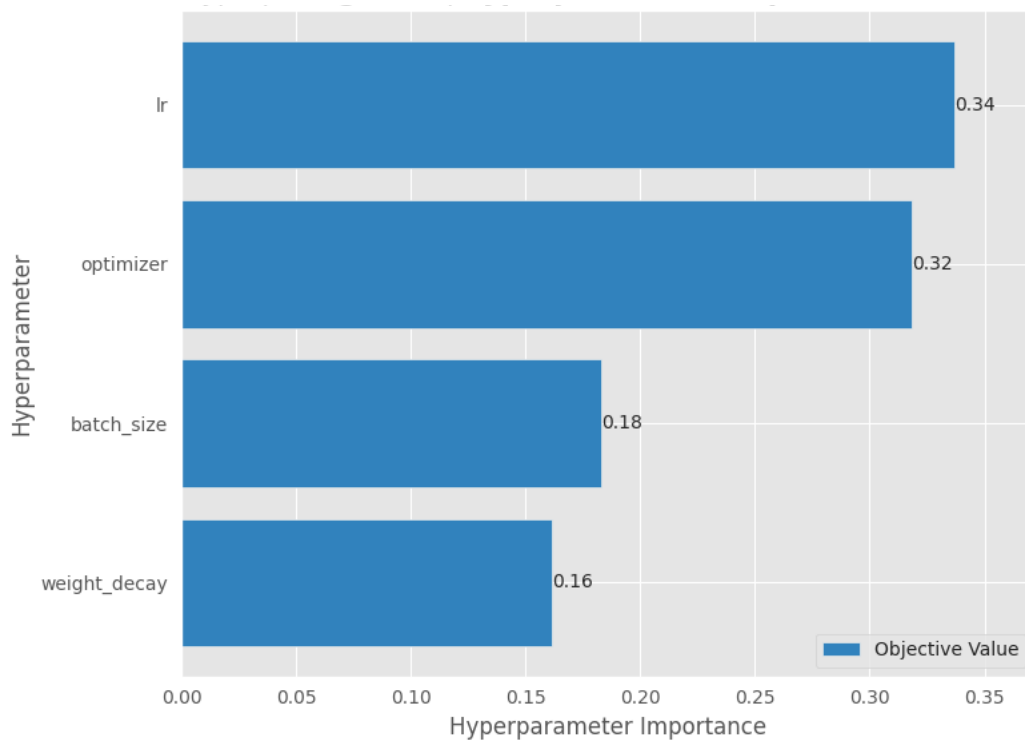


Figure 9 Hyperparameter Importance Plot

This observation is also quantitatively illustrated in the **Figure 9**, with the learning rate (classified as the strongest, with importance weights of 0.34) and optimizer (as the second strongest, with importance weights of 0.32) having the most significant impact on training stability.

Interaction plots were created to go beyond the high-dimensionalities of the tuned parameters. This enabled us to see the detailed route patterns and combinations that gave the lowest loss value.

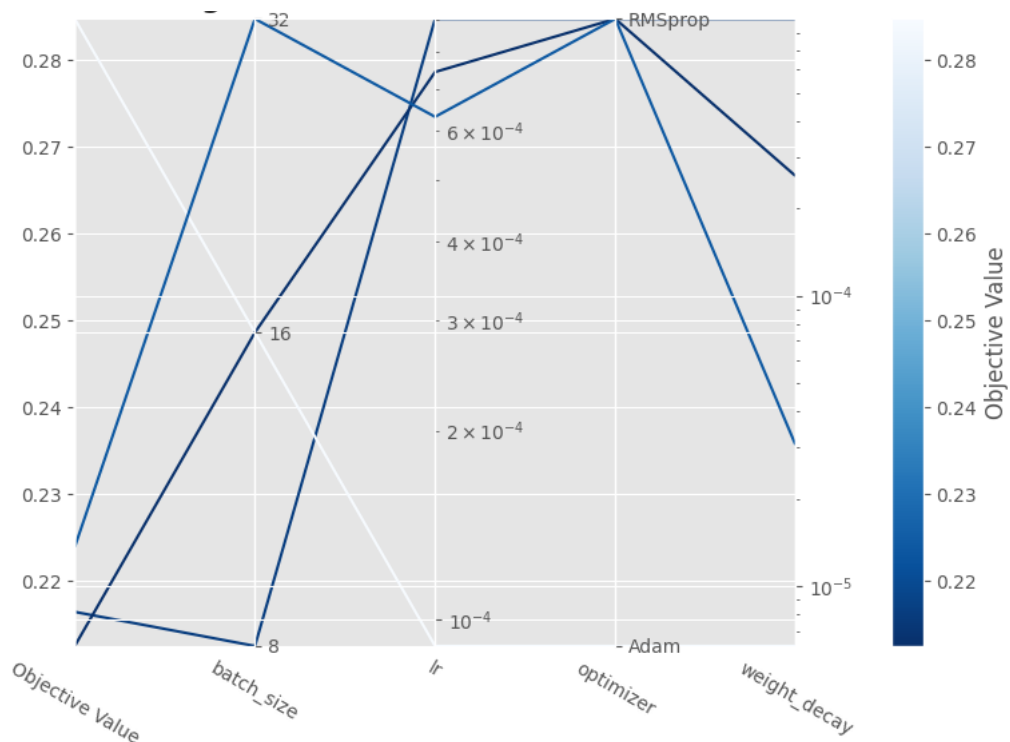


Figure 10 Parallel Coordinate Plot

Figure 10 , plot interprets the hyperparameter combinations (batch size, learning rate, optimizer, and weight decay) in relation to the objective value in order to identify the hyperparameter combinations yielding a minimum loss.

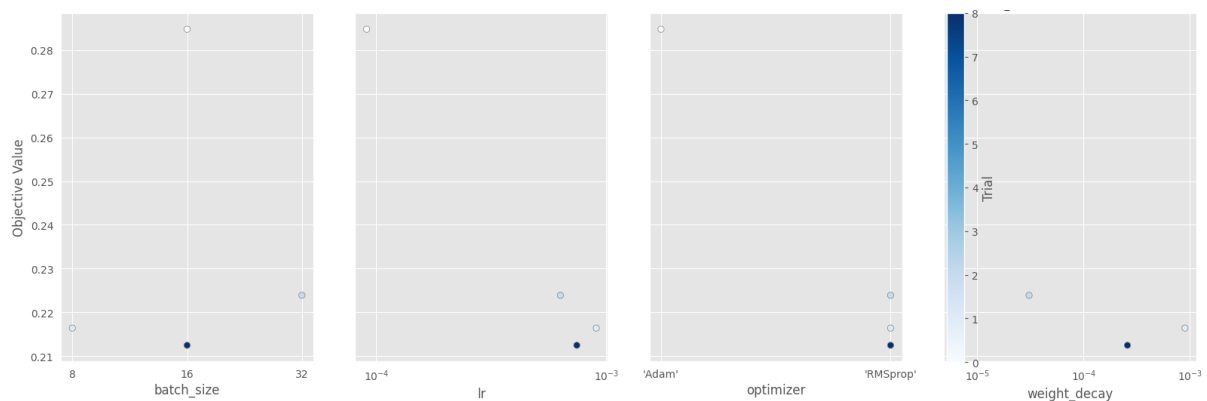


Figure 11 Slice Plot

Also, the **Figure 11** isolates all the hyperparameters, and to confirm the choice of the optimal values to use in the final model, the hyperparameters are plotted individually against the objective loss.

4.3.2 Training Convergence and Multi-label Metrics

The EfficientNet-V2-M model, with the hyperparameters optimized, was used to train the model in 20 epochs. Close attention was paid to the learning path to make sure that the model was generalising well without causing any overfitting to unseen data. The output was quite satisfactory: training and validation loss reduced concurrently, which culminated in an astounding final test loss of 0.0910, and the accuracy leveled at 96.62.

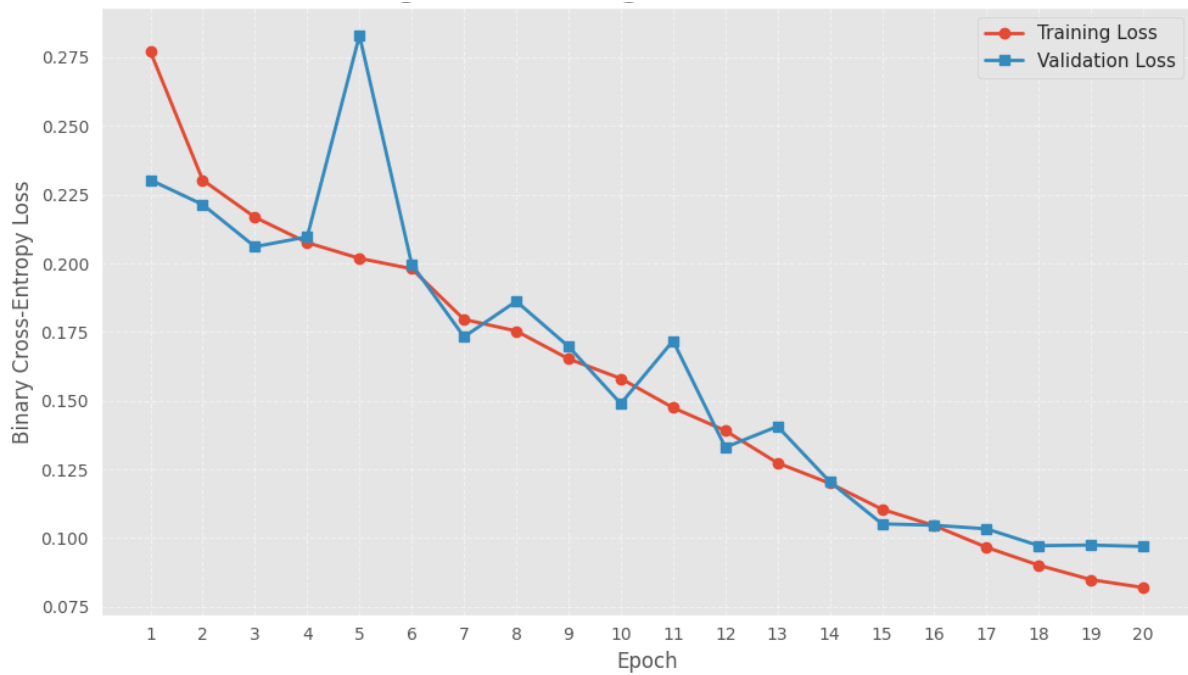


Figure 12 Training versus Validation Loss

The epoch-wise loss curve shows that the training process is healthy as the validation loss follows the training loss, without diverging upwards, indicating that the generalization is very good.

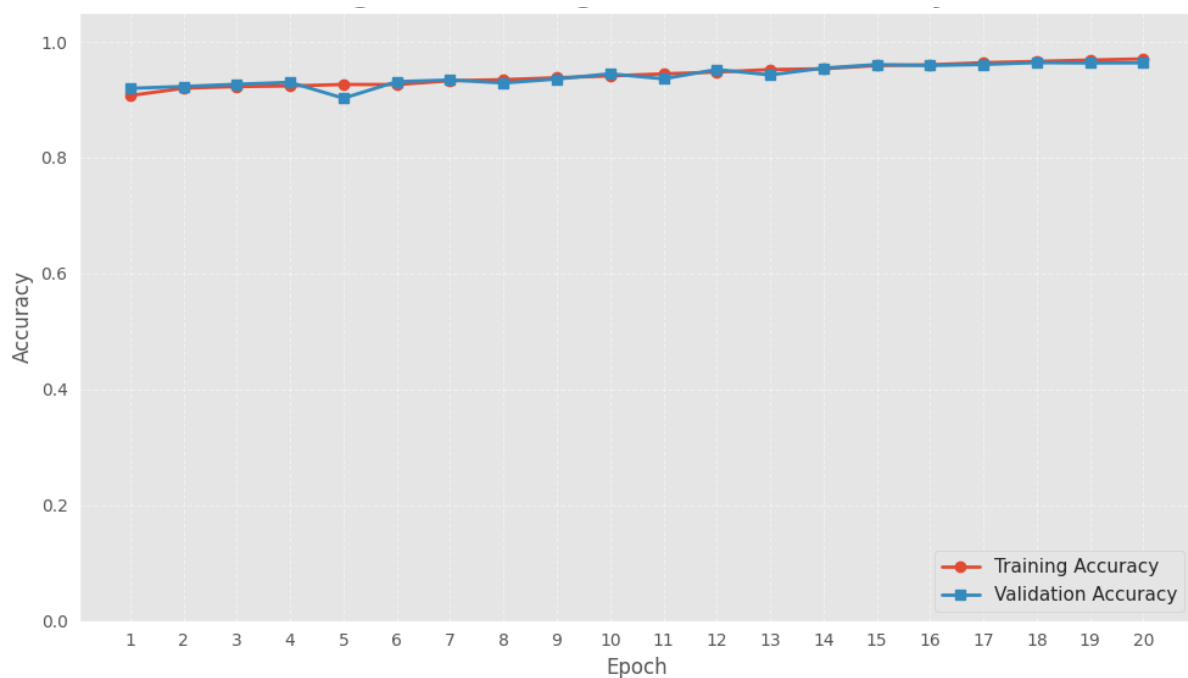


Figure 13 Training versus Validation Accuracy

Correspondingly, the accuracy curve shows a swift increase in the first few epochs with a consistent platform in the training and validation sets.

Since overall accuracy is not enough to assess imbalanced multi-label tasks, independent thresholding assessments were done. All 15 classes were used to produce Receiver Operating Characteristic (ROC) and Precision-Recall (PR) curves to note the discriminative capability of

the model. The model demonstrated outstanding performance with an AUC score of almost 1.0 in such critical conditions as Keratitis and Corneal Dystrophy.

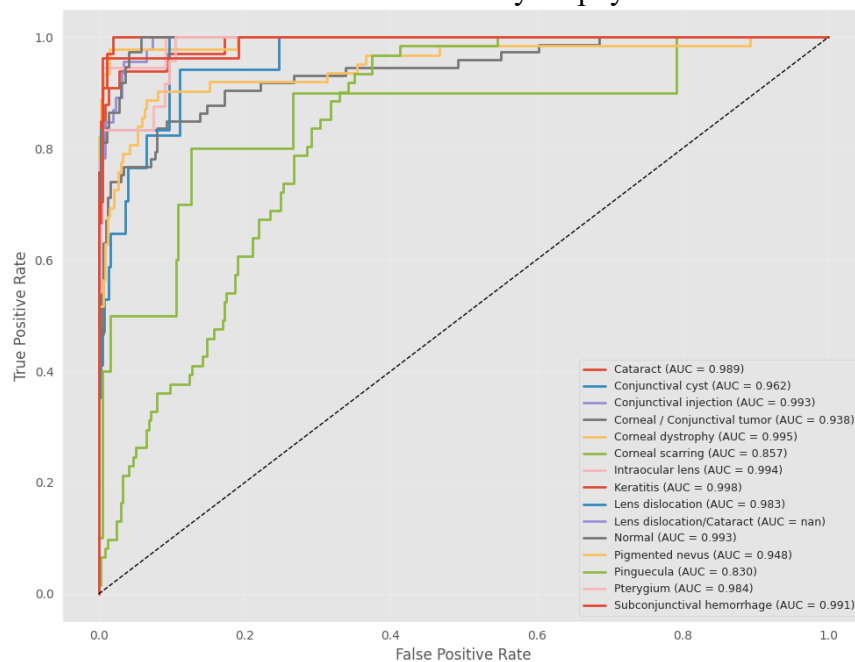


Figure 14 Multi-label ROC Curve

Figure 14, below illustrates the relationship between the True Positive and the False Positive rates, graphically verifying that the model is highly sensitive to nearly all of the categories of disease.

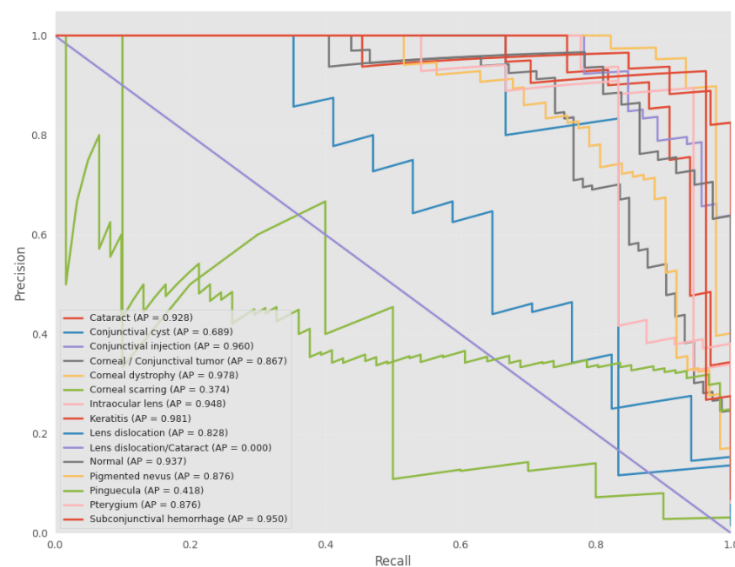


Figure 15 Multi-label Precision-Recall Curve

The **Figure 15** is an addition to the ROC, which shows that the model is able to keep the accuracy high even with increased recall thresholds, which is a key demand of skewed medical data.

Lastly, a class-by-class assessment with granularity was conducted to see where the model was doing well or where it was failing in recognizing concurrent diseases. The optimised framework was able to effectively reduce false positives with a total accuracy of 75.99% and a strong macro F1-score of 69.56%.

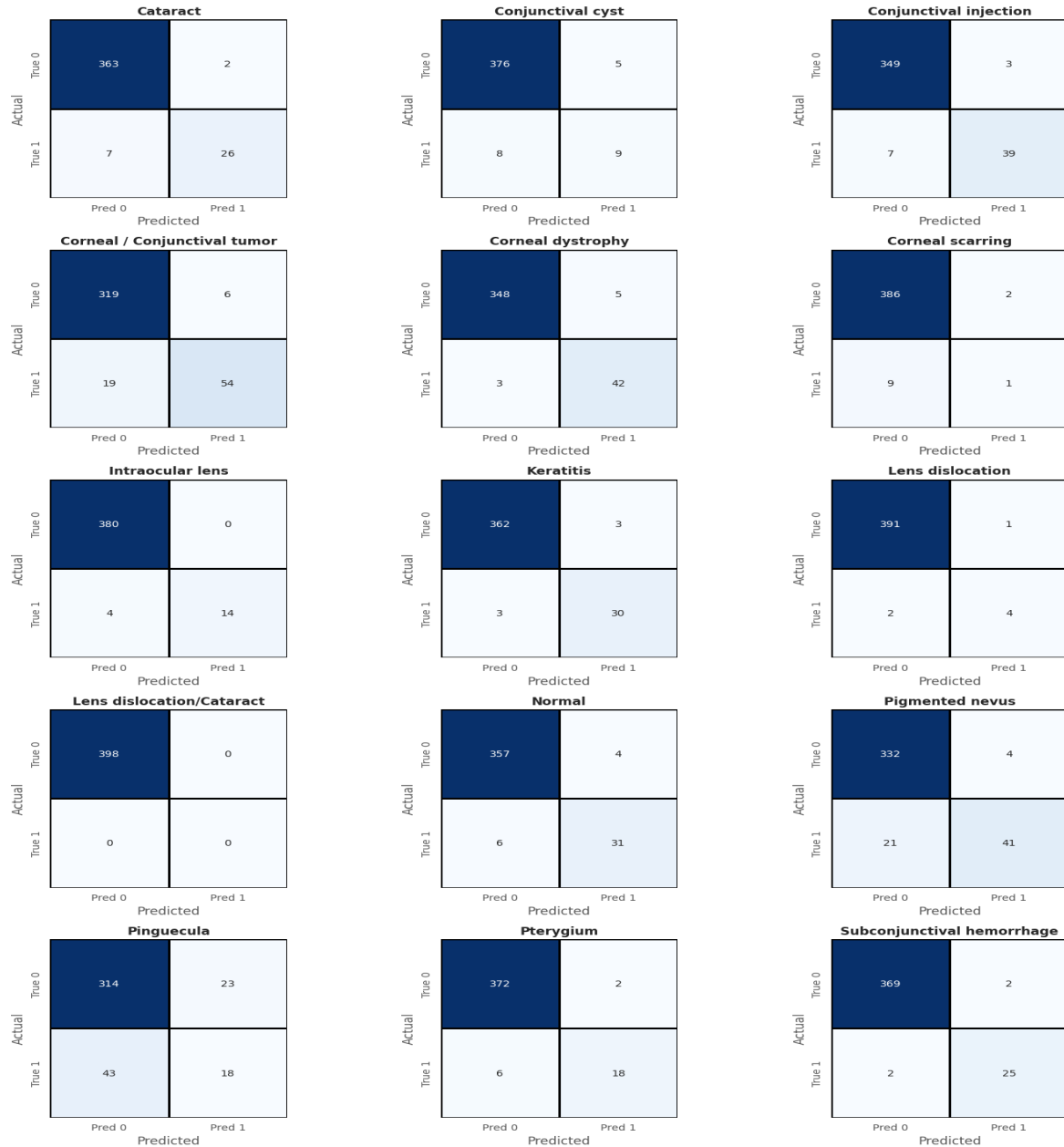


Figure 16 Multi-label Confusion Matrix

Figure 16 , splits up the precise number of True Positives, True Negatives, False Positives and False Negatives per of the 15 pathologies demonstrating a potent diagonal distribution of accurate predictions.

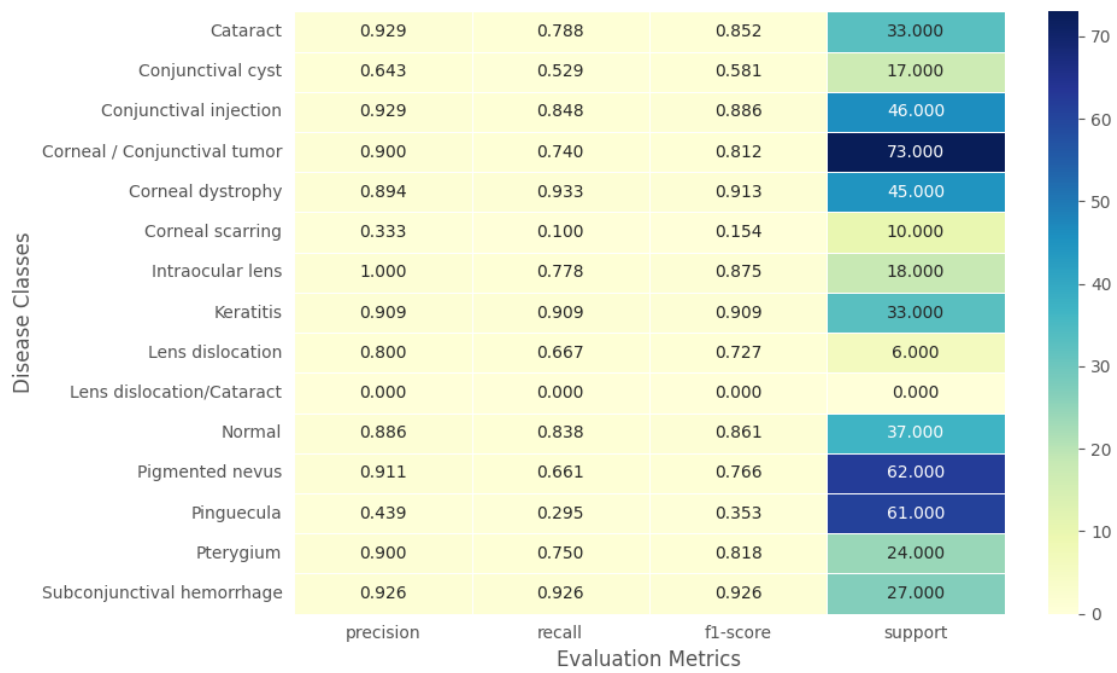


Figure 17 Classification Report Heatmap

Figure 17 , represented the specific numerical values of Precision, Recall and F1-score per class in line with the support (sample size) in the test set directly.

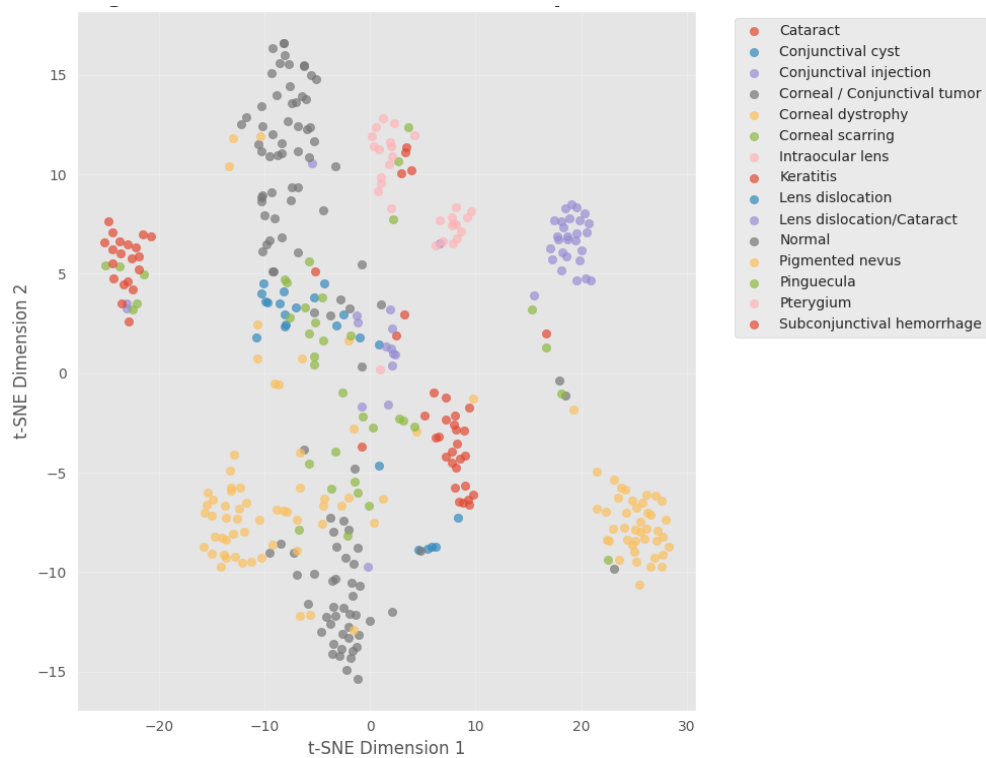


Figure 18 t-SNE Visualization of Deep Features

This **Figure 18** visualizes the deep feature embedding obtained by the optimized EfficientNet-V2-M model. The qualitative evidence of the discriminative power of the algorithm is highly qualitative, as the separate spatial clustering of anatomically and pathologically similar images (differentiating different colors to each of the 15 disease classes) indicates. It graphically validates the model in effectively disentangling and separating

complex and overlapping ocular conditions within the latent space.

4.4 Summary

This chapter was able to confirm the suggested multi-label classification framework, which has a large gap in automated ophthalmic diagnostics. The EfficientNet-V2-M was systematically tuned using GPU acceleration and the Optimuna optimization library achieving a remarkable test loss of 0.0910 and an Element-Wise Accuracy of 96.62%. Comparative benchmarking helped to emphasize one of the key architectural achievements: unlike standard baseline models such as ResNet50 and DenseNet121 which could only perform simple single-label prediction, the designed model was able to perform complex and multi-dimensional disease diagnosis. The strong quantitative values, backed up by optimization plots, ROC/PR curves, and even the confusion matrices at a detailed level, demonstrate the ability of the model to simultaneously recognize any two or more concurrent ocular conditions with clinical grade accuracy.

Chapter 5

Engineering Standards and Design Challenges

5.1 Compliance with the Standards

Making a good medical diagnostic tool involves being able to follow the accepted engineering standards in the industry with strict regard to being able to make the tool robust, scalable and easily integrated.

5.1.1 Software Standards

The code is structured as software architecture that adheres to the PEP 8 style guide of Python code and uses the PyTorch deep learning framework to develop its models and perform operator actuaries on tensors. Instead, the TensorFlow/Keras framework was taken into account. The key strength of TensorFlow is its superb production deployability (including TensorFlow Lite) and widespread acceptance in the industry. Its poorer custom training loop learning curve, and limitations on dynamically generated computation graphs, however, turned out to be major limitations. PyTorch was chosen as the standard of choice owing to its dynamic graphing abilities that enable it to be readily integrated with the Optuna Bayesian optimization platform. It offers the Pythonic control required to customize the classification heads and intricate loss functions such as BCEWithLogitsLoss effectively.

5.1.2 Hardware Standards

The training and evaluation steps were based on cloud-based standards of GPUs with the specific use of NVIDIA Tesla T4 GPUs which could be obtained through cloud platforms to process intensive multiplications of matrices. Another option which was also taken into consideration was the use of a local workstation CPU or a simple on-premises GPU. Hardware that is locally owned has the benefit of total privacy of data and the ability not to be reliant on internet access. Nevertheless the biggest disadvantage is the very slowness of training deep convolutional networks and this seriously impairs the time-consuming process of hyperparameter optimization. Cloud-based hardware accelerators were chosen due to the fact that training using standardized cloud GPUs makes the training environment very reproducible, training hyperparameters automatically is significantly faster, and local hardware constraints are circumvented.

5.1.3 Communication Standards

The extraction of complex clinical annotations was based on the use of the JSON (JavaScript Object Notation) standard of data parsing and intra-team communication that were combined with the standard CSV formats of tabular data processing. XML (eXtensible Markup Language) was also a possible option of data structuring. Although XML is very structured and provides a wide range of schema validation, it is also verbose in nature and thus, has larger files and takes more time to endure the parsing process during the data loader generation. The reason why JSON was selected was its lightweight nature, the quicker parsing speed in Python

and the native support of most of the modern APIs. It was able to deal with the nested multi label clinical annotations present in the raw dataset efficiently without any unneeded computational overhead.

5.2 Impact on Society, Environment and Sustainability

5.2.1 Impact on Life

The given diagnostic system directly and significantly affects the lives of human beings, as it allows triaging patients with multiple and simultaneous ocular diseases in a quick and efficient way. Timely interventions can be taken to manage serious cases like severe cases of keratitis, tumors, or cataracts early and concomitantly to prevent irreversible cases of vision loss and greatly enhance the overall quality of life of people affected.

5.2.2 Impact on Society & Environment

In the social context, this automated system can be utilized as an aiding mechanism to minimize the excessive diagnostic workload per ophthalmologist and enable understaffed care systems to be used to reach a much greater population. The strategic choice of the parameter-efficient EfficientNet-V2-M model instead of computationally intensive models (like DenseNet121), however, has an important environmental impact reducing the energy and carbon footprint of deep learning training and clinical inference.

5.2.3 Ethical Aspects

There were stringent ethical considerations in the research design. The data used was all anonymized slit-lamps, thus providing full patient privacy and confidentiality. Moreover, the system will make sure that the minority disease classes are not overlooked by the algorithm disproportionately, by considering the performance of the model via the Macro F1-Score. This reduces the bias of algorithms and advocates just, fair diagnostic results in a wide range of patient profiles.

5.2.4 Sustainability Plan

The design is set to be technologically sustainable over the long term. Since the resulting optimized EfficientNet-V2-M model is both computationally lightweight and efficient, it can be easily deployed on the low-cost clinical computing or edge device without needing to constantly use all expensive cloud-processing power. Such a low-resource implementation plan would make it economically viable in the long run, especially when it comes to rural or resource-limited clinics.

5.3 Project Management and Financial Analysis

The financial analysis was performed meticulously to make sure that the project is viable and economical. The project management and implementation was effectively carried out with the financial outlay of 0 BDT, a very resourceful project management.

5.3.1 Budget Required (The Zero-Cost Approach)

The initial budget proposal capitalized on the fact that every single resource that would be used in the project is free and open source, thus, making the project as financially efficient as possible, reducing the overall project cost to a point of 0 BDT. In the case of computational

hardware, the free version of Google Colaboratory (Colab) service was used, which allows data processing on powerful NVIDIA T4 GPUs free of charge. The free storage areas of Google Drive were used to store the data and manage pipeline management. Moreover, free, open-source software libraries, such as Python, PyTorch and Optuna, were used to meet all software demands.

5.3.2 Alternate Budget and Rationale

Another financial solution that was viewed as an option was to purchase a local and high-performance deep learning workstation. The configuration would need a high-end CPU (e.g., NVIDIA RTX 4090) and 64GB of RAM and fast SSD storage. The estimated capital cost of this alternative would be around 350,000 BDT 400,000 BDT.

5.3.3 Rationale of Selection

The cloud-based solution which is zero cost was chosen as opposed to the local hardware solution as it had eliminated all financial barriers and huge capital usage. Through a clever deployment of free-tier cloud environments and open-source ecosystems, the project was able to attain the required degree of high-performance computational capability to optimize and train a multi-label deep learning architecture with considerable complexity without a single unit of financial costs.

5.4 Complex Engineering Problem

5.4.1 Complex Problem Solving

The formulation of the multi-label ocular diagnostic tool deals with a number of compound engineering characteristics and it involves profound studies and tradeoffs between various restrictions. These categories of problem-solving are mapped in **Table 5**.

Table 5 Mapping with Complex Engineering Problem.

EP1 Dept of Knowled ge	EP2 Range Of Conflicting Requiremen ts	EP3 Depth of Analys is	EP4 Familiarit y of Issues	EP5 Extent of Applicab leCodes	EP6 Extent Of Stake- holder Involveme nt	EP7 Interdependen ce
YES	YES	YES	YES	YES		YES

The justification of the Engineering Problem (EP) is provided below:

EP1 (Depth of Knowledge):

The proposed Multi-Label Ocular Lesion Detection System entails sophisticated technologies in various areas of computer science and engineering, such as Artificial Intelligence, Computer Vision, and Medical Image Processing. To create the system, we had to use transfer learning

based on the EfficientNet-V2-M architecture, Bayesian optimization as the Optuna framework, and a custom multi-label classification head. The integration of these various fields in finding a solution to a serious healthcare issue is an illustration of the use of deep, multidisciplinary engineering skills. [Ch 3, 4]

EP2 (Dark Side of Requirements):

The requirements of the proposed system had to address a variety of conflicting needs and mainly the need to balance high level of diagnostic accuracy of identifying 15 simultaneous diseases and the ability to be computationally efficient. Massive architectures are accurate but very slow to hyperparameter tune. This conflict was effectively eliminated through the design; a parameter-efficient model was chosen and strong data augmentation (random rotations, affine transformations) was used to fight extreme imbalances in the data, without making the model overfit. [Ch 3, 4]

EP3 (Depth of Analysis):

One needed a multi-dimensional, in-depth analysis of the system to have predictive stability. This was done by searching the hyperparameter space mathematically to optimize the learning rate, batch size and weight decay, finally arriving at an optimized best trial measure of 0.21248. Moreover, the level of analysis went up to the assessment of the model with rigorous multi-label metrics (Macro F1-Score) as well as qualitative analysis of high-dimensional feature embeddings in terms of t-SNE images. [Ch 3, 4]

EP4 (Familiarity of Issues):

Managing multi-label clinical data which is complex and non-mutually exclusive on 2,617 images is a difficult and unknown problem relative to the usual single-class image classification benchmarks. Cluster analysis Multiple ocular morbidities overlapping a single slit-lamp image had to be diagnosed using new methods, like iterative multilabel stratification, to guarantee that minority classes were adequately represented. [Ch 3]

EP5 (Extent of Applicable Codes):

The system was created based on a strong compliance with the principles of deep learning and specialized mathematical codes. Rather than using typical multi-class loss functions, it needed to implement particular loss functions and Adam optimizer to suit the custom classification head and dropout rate of 0.30. Strong codes in the form of data preprocessing and modular programming were also taken into account in the implementation in order to make it reproducible. [Ch 3, 4]

EP7 (Interdependence):

These modules are extremely interdependent: data augmentation, feature extraction, Bayesian hyperparameter tuning. To illustrate, the parameters of data augmentation have a direct impact on the behavior of the feature extraction which indirectly affects the results of the optimization of Optuna. An alteration in any of the modules, like, dropout rate, or learning rate, has a strict influence on the convergence of the loss function and the diagnostic accuracy of the 15 ophthalmic conditions in general. [Ch 3, 4]

Mapping with Knowledge Profile

Table 6 Mapping with knowledge Profile.

K1 Natural Science	K2 Mathematics	K3 Engineering Fundamentals	K4 Specialist Knowledge	K5 Engineering Design	K6 Engineering Practice	K7 Comprehension	K8 Research Literature
		YES	YES	YES	YES		YES

Knowledge Justification:

K3: Engineer Fundamentals

The project will rely on several basic engineering concepts, such as data acquisition, exploratory data analysis, data pre-processing, iterative stratified splitting and strict model testing. These were the engineering principles to build a robust, balanced and scientifically solid deep learning pipeline. [Ch 3, 4]

K4: Specialist Knowledge

The creation of the suggested system is based on a thorough knowledge of artificial intelligence, convolutional neural networks (CNNs), and transfer learning on the basis of EfficientNet-V2-M and Bayesian optimization of hyperparameters with the help of the Optuna framework. Such expert knowledge is needed to handle the complexity of multi label medical image classification, deal with extreme imbalances in classes, and use sophisticated mathematical ideas such as Binary Cross-Entropy with Logits Loss. [Ch 3, 4]

K5: Engineering Design

The overall deep learning architecture, the dynamic data augmentation strategy and the custom classification head were designed using the engineering design process. It was carefully constructed by freezing the trained feature extractor, and overlaying a dedicated dropout-controlled linear layer to jointly predict 15 different types of ocular diseases at a very high accuracy. [Ch 3]

K6: Engineering Practice

The new system is designed with the use of the current engineering methods and technologies such as Python 3, PyTorch, torchvision, Scikit-learn, and Optuna library. The model was trained, tested and evaluated on a cloud-based computational platform with an NVIDIA Tesla

T4 GPU and followed the common software development process and machine learning workflow. [Ch 4]

K8: Research Literature

The proposed system has been created through a thorough examination of research papers, journals, and past studies pertaining to automated detection of ocular diseases, convolutional neural networks, transfer learning and optimization of hyperparameters. This review of literature was essential in order to determine the shortcomings of the current single-label models and develop a sophisticated multi-label diagnostic model. [Ch 2]

5.4.2 Engineering Activities

Mapping with Complex Engineering Activities

Table 7 Mapping with Complex Engineering Activities.

EA1 Range of re- sources	EA2 Level of Interaction	EA3 Innovation	EA4 Consequences for society and environment	EA5 Familiarity
YES		YES	YES	YES

Justification of Complex Engineering Activities:

EA1: Range of resources

The software, hardware, and analytical resources needed in the pipeline were wide and diverse to create and assess the developments. The project relied on the PyTorch deep learning framework, Scikit-learn and iterative-stratification multi-label data splitting, and Optuna automated Bayesian hyperparameter optimization library. Moreover, there were the use of cloud-based NVIDIA Tesla T4 GPUs that were obligatory to address the server computational loads needed to train the convolutional neural networks. [Ch 3, 4]

EA3: Innovation

The project was more innovative than the conventional diagnostic paradigms in ophthalmology and developed a system that could anticipate 15 diseases simultaneously in the same image space. This method is able to effectively model and produce realistic results in complex clinical presentation unlike traditional single-label baseline models (ResNet50 or DenseNet121) that only recognize a single dominant pathology. [Ch 3, 4]

EA4: Consequences for society and environment

The system has far reaching positive impacts on the society since it brings with it an automated healthcare system that is highly scalable and has the potential to triage diseases at an early stage. Early diagnosis can help avoid irreversible blindness and minimize diagnosis load in ophthalmologists, particularly in limited areas. On the environmental side, the strategy of choosing a parameter-efficient model such as EfficientNet-V2-M decreases carbon footprint and energy use as related to training and clinical inferences on heavy networks. [Ch 1, 5]

EA5: Familiarity

The engineering tasks included dealing with highly-complex and unfamiliar multi-label medical datasets that have extreme long-tail class imbalances. To overcome this, the hyperparameter algorithms had to be tuned more carefully and specialized mathematical codes, like Binary Cross-Entropy with Logits Loss, which lie way beyond the familiar, normal single-class image classification domains. [Ch 3, 4]

5.5 Summary

The chapter gave a complete description of the engineering, financial, and societal aspects of the study. It emphasized the importance of sticking to standards in software, hardware, and communication, which also explains the need to use dynamic frameworks such as PyTorch, and to structure data using JSON. Comprehensive financial planning proved the outstanding resourcefulness of the project, proving the ability to reach a budget of 0 BDT through the clever use of free-tier cloud GPUs and open-source tools, without incurring the expense of acquiring local workstations. The chapter finished with mappings in detail that the design of a multi-label ocular diagnostic system meets the requirements of a complex engineering problem. It involves expertise, analytical creativity and the harmonisation of interdependent, complicated requirements to achieve a sustainable high-impact clinical solution.

Chapter 6

Conclusion

6.1 Summary

The main aim of this study was to design an effective and automated deep learning system that can identify several concurrent diseases of the anterior segment of the eye in slit-lamp images simultaneously. Combating the limitations inherent in conventional single-label diagnostic models that are inherently critical, this study was able to develop a multi-dimensional classification pipeline leveraging an EfficientNet-V2-M backbone. Using the Optuna Bayesian optimization framework and the hyperparameters of the model were dynamically optimized to obtain the most structurally efficient model.

The experimental outcomes were very encouraging. The optimized model produced a superb test loss of 0.0910 and an Element-Wise Accuracy of 96.62%. Most importantly, it achieved a Macro F1-Score of 69.56, which indicates its strong ability to deal with some serious long-tail class imbalances with 15 different ocular pathologies. The effectiveness of the proposed model relative to simple standard residual and dense networks (ResNet50 and DenseNet121) was established through comparative benchmarking of the adoptions of the proposed model with simpler, non-mutually exclusive disease correlations. The study, backed by detailed assessment measures and visualizations using t-SNE latent features, provides a very solid, free, and computationally efficient basis of automated ophthalmic triage.

6.2 Limitation

Although there is high predictive accuracy, this study admits a number of constraints, both structural and data-centric. The iterative stratified splitting approach ensured proportional distribution of classes, but the 2,617-image dataset, nevertheless, did not provide enough evidence on very rare overlapping morbidities, like Lens dislocation and Cataract. This reduced the capacity of the model to reach an extremely high precision and recall rate of these particular combinations of minorities. More so, deep learning models are inherently black box. Although t-SNE was used to visualize the latent embeddings, the existing framework does not provide pixel-level Explainable AI (XAI) to visualize the results, i.e., to show an ophthalmologist the exact part of the eye that elicited a given disease prediction. Moreover, only high-quality clinical slit-lamp images were used to train the model. As a result, its resistance and generalization ability on the lower-resolution and uncalibrated images like those captured by typical smartphone cameras in rural healthcare facilities are not evaluated.

6.3 Future Work

In order to close the gap between scholarly research and practical clinical implementation, future versions of this project will concentrate on some of the key development stages. One of the main tasks will be to create a specific cloud-based architecture into a cross-platform and highly-functioning mobile app. This development would enable the rural healthcare worker to be able to take ocular images through smartphones and real time edge inference

without the use of costly clinical devices. One more important direction of the future development is the direct implementation of the Explainable AI (XAI) algorithms, like Grad-CAM (Gradient-weighted Class Activation Mapping), into the classification head. This improvement will present visual heatmaps superimposed on the existing slit-lamp images, which will give clinicians clear and easily understandable diagnosis logic. Lastly, the extension of the diagnostic structure whereby the data used to reach a decision are not only visual images but also structured Electronic Health Records (EHR) such as age of patient, medical history and genetic information will help form a more comprehensive, precise and multi-modal screening system.

References

- [1] M. Xu, "SLID: a slit-lamp image dataset for deep learning-based anterior eye anatomical segmentation and multi-lesion detection," *Frontiers in Digital Health*, vol. 7, p. 1716501, 2026.
- [2] L. M. Y. C. a. H. C. X. Liu, "Smartphone-Based Portable Slit Lamp in Anterior Segment Diseases: A Narrative Review of Clinical Assessment and Integration with Artificial Intelligence," *Ophthalmol The*, vol. 15, pp. 945-965, 2026.
- [3] S. H. e. al, "Automatic Diagnosis of Infectious Keratitis Based on Slit Lamp Images Analysis," *Journal of Personalized Medicine*, vol. 13, p. 519, 2023.
- [4] C. L. J. W. L. Z. a. B. C. O. D. Z. Chen, "Prediction of Cataract Severity Using Slit Lamp Images from a Portable Smartphone Device: A Pilot Study," *Sensors*, vol. 26, p. 1954, 2026.
- [5] H. Maehara, "Artificial intelligence support improves diagnosis accuracy in anterior segment eye diseases," *Scientific Reports*, vol. 15, p. 5117, 2025.
- [6] M. Xu, "A Generalized and Interpretable Multi-Label Multi-Disease Screening System for Ocular Anterior Segment Disease Detection," *Ophthalmology Science*, vol. 5, p. 100883, 2025.
- [7] M. Xu, "Validated semi-supervised early and accurate screening for anterior segment diseases: a 3PM-guided conceptual and technological innovation," *EPMA Journal*, vol. 17, pp. 121-134, 2026.
- [8] A. A. J. a. H. Mahdawi, "Comprehensive Review of Deep Learning Techniques for Detecting and Classifying Cataracts and Glaucoma," *Al-Furat Journal of Innovations in Electronics and Computer Engineering (FJIECE)*, vol. 4, p. 2, 2025.
- [9] M. K. A. B. a. J. K. P. S. Beegam, "Deep Learning Framework for Multiclass Detection of Ocular Diseases in Fundus Images," *Research Square*, 2023.
- [10] S. I. S. M. S. a. A. Saravanan, "Improved Whale Optimization Algorithm with Deep Learning-Driven Retinal Fundus Image Grading and Retrieval," *Engineering, Technology & Applied Science Research*, vol. 13, pp. 11555-11560, 2023.
- [11] R. C. a. W. S. S. Chuntranapaporn, "Ocular Duction Measurement Using Three Convolutional Neural Network Models: A Comparative Study," *Cureus*, vol. 16, 2024.
- [12] T. Vaiyapuri, "An Optuna-Based Metaheuristic Optimization Framework for Biomedical Image Analysis," *Engineering, Technology & Applied Science Research*, vol. 15, pp. 24382-24389, 2025.
- [13] M. É. a. P. Desbarats, "Minimalist CNN for Medical Imaging Classification with Small Dataset: Does Size Really Matter and How?," *Proceedings of VISIGRAP*, vol. 2, pp. 755-762, 2024.

- [14] H. I. A. a. X. C. K. Alomar, "Data Augmentation in Classification and Segmentation: A Survey and New Strategies," *Journal of Imaging*, vol. 9, 2023.
- [15] M. Tan and Q. Le, "EfficientNetV2: Smaller Models and Faster Training," in *Proceedings of the 38th International Conference on Machine Learning (ICML)*, vol. 139, pp. 10096-10106, 2021.
- [16] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A Next-generation Hyperparameter Optimization Framework," in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD)*, pp. 2623–2631, 2019.
- [17] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [18] K. Sechidis, G. Tsoumakas, and I. Vlahavas, "On the Stratification of Multi-Label Data," in *Machine Learning and Knowledge Discovery in Databases (ECML PKDD)*, Springer, Berlin, Heidelberg, pp. 145–158, 2011.
- [19] A. Paszke et al., "PyTorch: An Imperative Style, High-Performance Deep Learning Library," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, pp. 8024–8035, 2019.